The New Standard in Precision Bolt Tensioning Solutions

AUSTRALIAN MADE

Safety Disclaimer & Warranty Information

SAFETY

Before attempting installation of Technofast products, it is imperative that operators understand the installation and tensioning procedures contained in the manual supplied with the product. **DO NOT ALLOW UNTRAINED STAFF TO ATTEMPT INSTALLATIONS.** Do not allow personnel into the installation area without first instructing them on the safety hazards of hydraulic components.

Pre-Installation Safety Checklist:

1. Are all persons in the installation area aware of safety procedures contained in the bolt tensioning product manual?
2. Have all persons in the installation area been issued with adequate eye and hearing protection devices?
3. Have all persons in the installation area been issued with suitable protective and hi-vis clothing?
4. Are installation personnel trained in the dangers of misuse of high-pressure hydraulic equipment?
5. Is the installation area free of unnecessary obstructions which could endanger personnel performing assigned tasks?
6. No person should be allowed in close proximity to hydraulic components while pressurising the system.
7. When pressurising the system, the pump operator must remain at the station in case of emergency.
8. Set air regulator to achieve the correct oil pressure for the product before coupling and tensioning begins.

REMEMBER: Fluids under high pressure can cause damage or injury if the device is improperly used.
Incorrect installation procedures can lead to major safety problems and void warranty.

Important Information Before Begining Installation:

- Operators should wear eye protection and observe safety practices applicable to the use of high-pressure pumping equipment.
- Do not advance components over their recommended tolerances or damage will occur.
- If the gap between the bolt tensioning unit and equipment is excessive, the unit has likely not been installed sufficiently. Release any pressure immediately and repeat the operating procedure as per the operation manual.
- Determine the required pressure (listed on the top of the Technofast products or on the product drawing or certificate supplied with the bolt tensioning unit) and ensure the maximum stroke is checked where applicable.
- Determine the pump and hose configuration and ensure all equipment is installed correctly and ready to operate.
- Regulate the pump to achieve the required pressure, ensuring a steady increase to avoid sudden pressure surges or bursts.
- Ensure the mating spot-face is greater in diameter than the spherical washer and is clean of paint, foreign matter, swarf, or weld spatter.
- Ensure that the base or washer face of the bolt tensioning unit contacts the joint face evenly and is fully supported across its full diameter.

Removal of Bolt Tensioning Unit:

To remove the bolt tensioning unit after operation, please follow the procedures as directed in the product operation manual provided with the unit. Operators must wear eye protection and observe high-pressure safety practices.

Servicing and Maintenance:

Servicing Technofast products at regular intervals is crucial to maintain operation. It allows for preventive maintenance, ensuring potential issues are identified and resolved before they escalate. Regular servicing includes lubrication, inspection, and detection of damage or wear, optimising performance and preventing unexpected failures. Adhering to recommended intervals and procedures ensures continuous operation and reliability.Refer to the product’s operation manual for more details.

WARRANTY AND LIABILITY

Technofast products and related intellectual property are covered by Australian and International Patents. Due to the nature of applications, no responsibility for performance can be accepted by the manufacturer beyond the terms of the Limited Warranty. Technofast products are warranted free of original defects in material and workmanship for a period of one year from the date of shipment to the first user. This warranty is void if the product has been disassembled in an unauthorised facility, if substitute parts have been used, or if the product has been used in any manner not in accordance with manufacturer directions. Technofast’s obligation is limited to replacement, and in no event shall liability be accepted for any loss or damage, consequential or special, arising in connection with the use of these products.

About Us

Technofast, Crestemad QLD

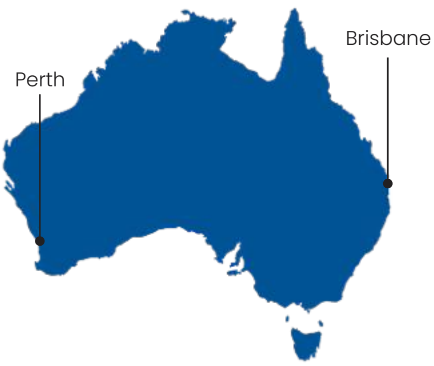
Technofast has been providing innovative and economical bolt tensioning solutions to industry around the world from its Australian base for over thirty years. The company's products utilise hydraulic power for applying the high forces needed to correctly tension bolts and shafts, and have been fitted in a wide variety of applications.

Our Network

With decades of Australian manufacturing expertise Technofast delivers reliable, world-class bolting solutions that keep critical industrial operations running smoothly worldwide.

Brisbane, QLD: Technofast Head Office and manufacturing facility is located within the Crestmead Industrial Estate, situated in the Southern Corridor. This site supports the design, manufacture, assembly and testing of the company's products. Technofast also maintains dedicated facilities for extensive research and development activities.

Perth, WA: Technofast's Service Division operates from the industrial Rockingham precinct, specialising in comprehensive on-site support for bolt tensioning, precision machining, and infrastructure maintenance. Serving leading industrial clients across Australia and Asia, this regional hub provides technical excellence through custom-engineered installations, technical analysis, and safety training. The facility also maintains a specialised equipment fleet available for short-term or extended hire



America (Tampa, Florida & Mooresville, North Carolina): Technofast's Americas division delivers customised, precision hydraulic bolting solutions directly to clients across the United States, Canada, and South America. Backed by Technofast's core engineering expertise, our American team specialises in high-temperature technology & custom-engineered systems tailored for demanding mining, oil and gas, and nuclear power sectors.

Industries Applications

Our tooling range serves a wide variety of industries, including, but not limited to:

Manufacturing

All components are produced using modern CNC machines, ensuring that exceptional quality is maintained throughout the manufacturing process.

Processes include:

- CNC Milling
- CNC Turning
- Laser Engraving

Certified alloy steels are typically used for these to ensure toughness and durability. Products can also be finished with a variety of surface treatments to suit environmental conditions.

Design

Our engineering staff utilise state of the art drawing and 3D modelling packages to design Technofast's unique products. This digital approach to design allows the virtual models to be effectively assessed for performance before any prototypes are made and tested.

The method is efficient, allowing significant reductions in overall size of components, and economical as significant cost reductions can be achieved, resulting in savings for clients.




Sales & Aftermarket Support

Technofast has an extensive network of Distributors and Representatives within Australia and internationally who are trained to assist with enquiries, installations and product support. Our staff provide assistance with training and consultation with customers. All goods produced by Technofast are warranted against failure due to design, materials, or manufacturing faults. Details are available from Technofast's Warranty Policy.

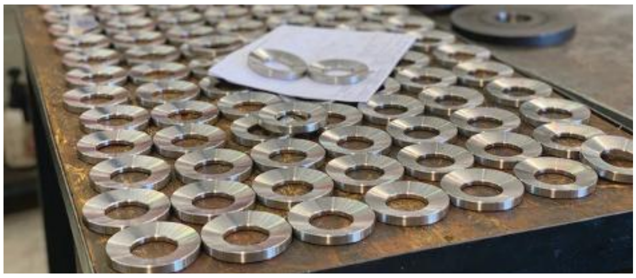
Testing

All finished products are rigorously tested prior to despatch, entailing physical checking for conformity of finish, final measurement, and performance under load.


For many of the company's innovative solutions, a full R&D program may be undertaken to take a concept to performance prior to full scale manufacture.

This may include:

- Prototyping
- Cycle Testing
- Full Load and Overload Testing of Components
- Pressure Testing
- Failure Analysis

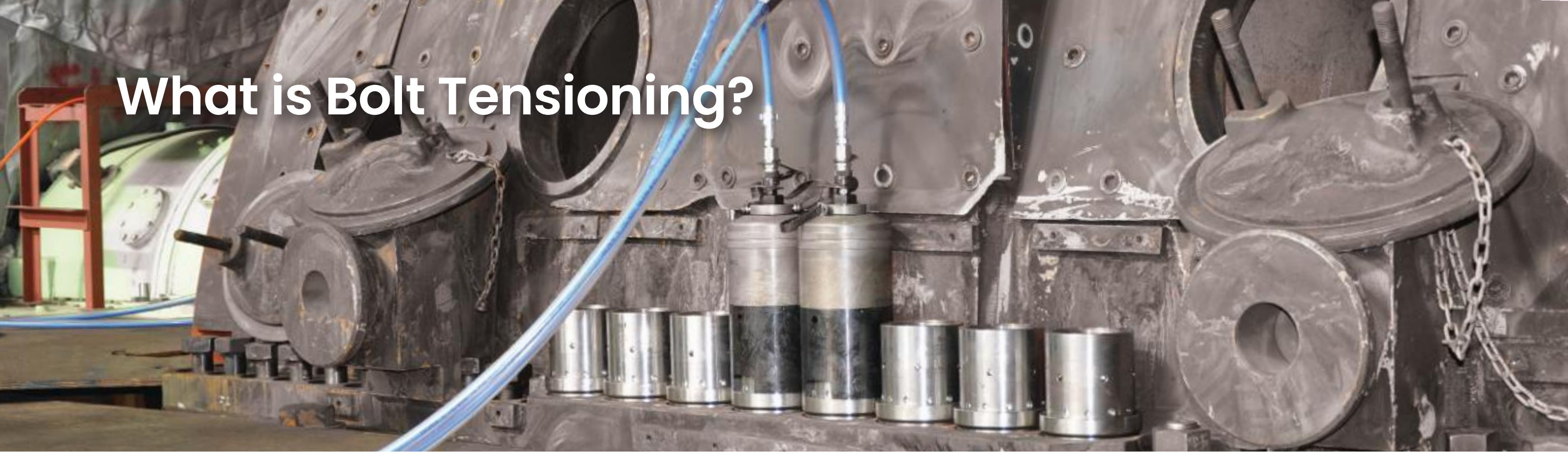



Quality Assurance

Technofast has a well-proven ISO 9001 compliant Quality Management System in place to ensure that exceptional product quality is maintained.

As a proud Australian manufacturer, our products are designed and built at our Brisbane headquarters, supporting industries across Australia as well as Europe, Asia, Africa, North America and South America.

Our commitment to Australian-made bolting and tensioning solutions ensures that we deliver innovative, world-class technology with shorter delivery times and higher reliability for our clients.



What is Bolt Tensioning?

A well-designed bolted joint relies on the natural elasticity of steel to resist the forces of separation and shear.

Because a bolt wants to return to its original length after being stretched, it creates the force needed to clamp joint faces together and lock them with friction.

While there are several ways to achieve this 'stretch,' the precision of that tension is what ultimately determines the strength and safety of the entire assembly.

Understanding Bolt Function

There are three primary ways to stretch a bolt and create a secure joint.

While each method results in a clamping force, they differ significantly in terms of accuracy, ease of use, and the equipment required.

Torque Method (Twisting)

Torque uses screw threads to provide a mechanical advantage.

When a lever or wrench is used to rotate a nut against the joint face, that turning force (torque) stretches the bolt as it moves through the screw path.

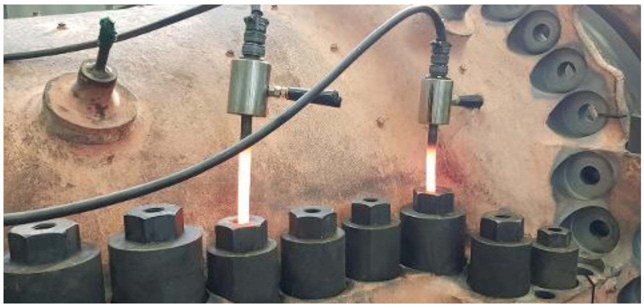
While being a commonly used method, it can be inaccurate due to the "vagaries of friction", the rubbing between the threads and the nut, making it difficult to tell how much force is actually stretching the bolt versus just fighting the friction.

Thermal Elongation (Heat)

This involves heating a bolt so it expands, putting it into place, and then allowing it to cool.

Because the bolt naturally wants to return to its original "free" length as it cools, it creates a clamping force against the retaining nut.

While effective, this is generally a last resort for difficult cases.



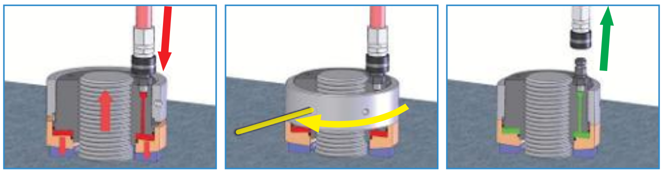
Direct Tension Method (Hydraulic)

For the highest level of accuracy, equipment like Hydraulic Nuts and Bolt Tensioners are used.

These tools are designed with a simple sealed internal chamber (an annulus) that is pressurised with fluid from an external source.

When attached to the bolt, this applied pressure generates a straight, axial force in the shank.

Once a specific pressure is reached, the locking mechanism is engaged and the pressure is released, perfectly transferring the hydraulic force into a secure mechanical hold.



Why use Direct Tension?

- Safe
- Accurate
- Calculated
- Fast
- Versatile
- Reliable
- User Friendly

Approved Distributors

Technofast Industries are approved Distributors of both CEJN and SITEC Products within Australia, and recommend the use of these products for all high pressure connection of Bolt Tensioning tools and equipment.

CEJN

Technofast Industries are authorised distributors of CEJN products within Australia.

For all high-pressure connections on bolt tensioning tools and equipment, we recommend and supply this range to ensure correct product compatibility and performance.

Operating since 1955, CEJN designs quick-connect couplings and nipples based on small external dimensions and large internal ones.

This intelligent technical layout ensures their components deliver high flow capacity while remaining robust and lightweight.

By integrating these high-performance components into our specialised hydraulic setups, we provide reliable, efficient systems engineered for demanding industrial workloads.

SITEC

SITEC provides a wide range of precision engineered extreme pressure tubing and fittings internationally from the company's Switzerland factories.

These products are suited to applications requiring durable and reliable hydraulic connections between components.

Typical uses are those for laboratories, process control mechanisms, and testing equipment.

The high quality Stainless Steel construction enables SITEC products to be used in corrosion prone areas, and those where resistance to chemical attack is necessary.

Technofast Industries proudly represent SITEC, and can offer a complete consulting service for products within their range to ensure optimum performance and cost effectiveness for every application.

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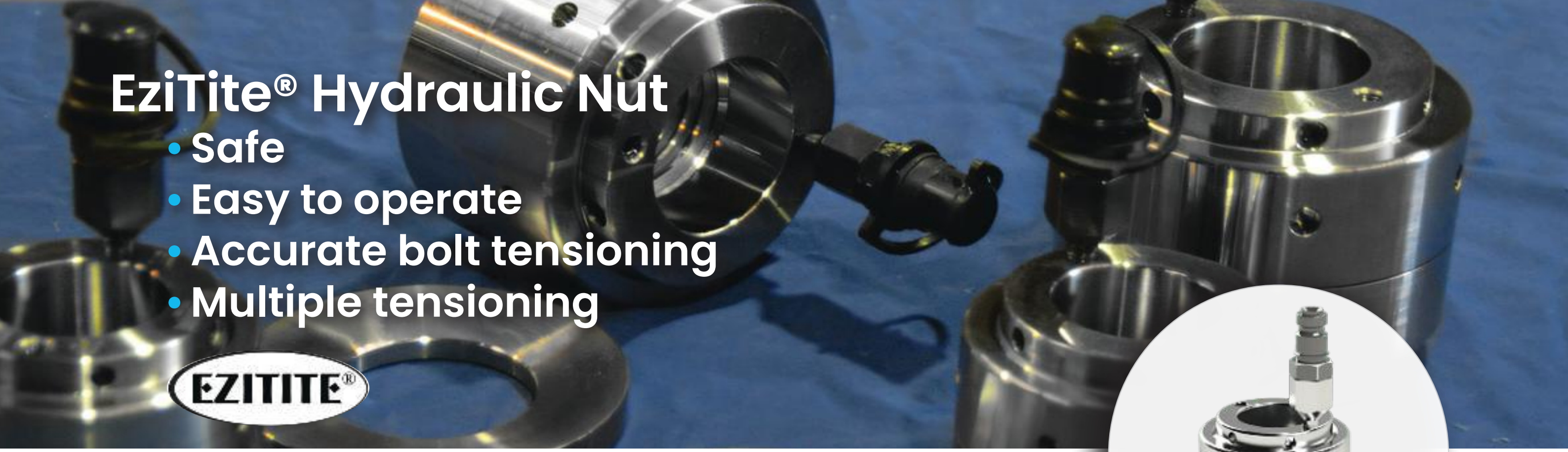
Bolt Tensioning Solutions



EziTite® Hydraulic Nut

- Safe
- Easy to operate
- Accurate bolt tensioning
- Multiple tensioning

EZITITE®

The **EziTite® Hydraulic Nut** is a precision engineered, high pressure, high performance, hydraulically operated bolt tensioning device.

It can be quickly and easily fitted when used with Technofast pumping equipment (e.g. Hand Pumps and High Pressure Air and Electric Hydraulic Pumps).

The **EziTite® Hydraulic Nut** is manufactured in a choice of alloy or stainless steel to suit the required application.

Operation

The **EziTite® Hydraulic Nut** assembly is screwed by hand onto the bolt (replacing the original Nut) until the base is tight against the working face.

Hydraulic pressure is then applied through the nipple fitting on top of the nut body into the sealed chamber, forcing the piston and the nut body apart, thus stretching and tensioning the bolt through the joint.

The threaded lockring, mounted on the nut body, is then screwed against the abutting face to retain the induced load in the bolt.

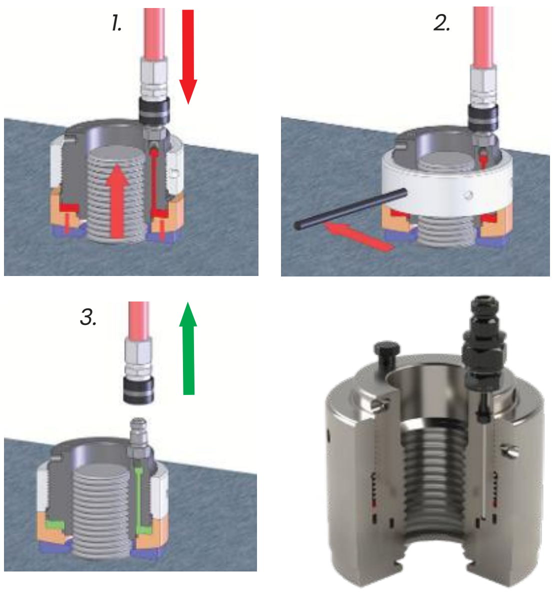
The pressure is then simply released and the hydraulic coupling reversed from the nipple fitting to complete the operation.

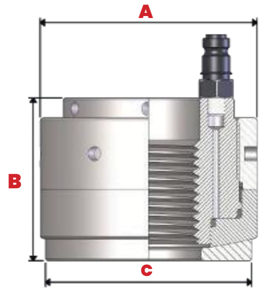
Features

The patented design of the **EziTite® Hydraulic Nut** has given it improved technical efficiency of operation and cost/benefit advantages over other tensioning devices.

The **EziTite® Hydraulic Nut** provides advantages such as:

- Reduced maintenance down time
- Improved safety on the job
- Reliable and precise tensioning
- Is user-friendly
- Fast to fit and remove





Specifications

COMMERCIAL GRADE : Max. operating pressure 700 Bar

Model Part #	Thread Size	Force (kN)	(A) Nut OD (mm)	(B) Height short Stroke (mm)	(B) Height Long Stroke (mm)	Short Stroke (mm)	Long Stroke (mm)	(C) Washer Diameter	Weight Approx. Short (kg)	Weight Approx. Long (kg)
EZI-CM24X3	M24	57	Ø58	61	73	6	12	Ø52	1	1.21
EZI-CI100U	1	64	Ø58	61	73	6	12	Ø52	0.99	1.19
EZI-CI102U	1.125	80	Ø64	63.5	75.8	6	12	Ø58	1.22	1.46
EZI-CM30X35	M30	86	Ø68	66.2	78.2	6	12	Ø62	1.48	1.72
EZI-CI104U	1.25	106	Ø72	65.5	77.5	6	12	Ø68	1.62	1.91
EZI-CI106U	1.375	119	Ø78	68	80	6	12	Ø74	1.97	2.31
EZI-CM36X4	M36	135	Ø78	68	80	6	12	Ø74	1.91	2.27
EZI-CI108U	1.5	143	Ø82	69.7	81	6	12	Ø78	2.17	2.56
EZI-CI110U	1.625	177	Ø88	72.5	84.5	6	12	Ø82	2.57	3.01
EZI-CM42X45	M42	177	Ø88	72.5	84.5	6	12	Ø82	2.54	2.99
EZI-CI112U	1.75	186	Ø94	79.2	102.2	8	20	Ø86	3.21	4.17
EZI-CM48X5	M48	234	Ø104	80.2	103.2	8	20	Ø98	4	5.18
EZI-CI200U	2	245	Ø108	86.9	108.9	8	20	Ø102	4.62	5.83
EZI-CM56X55	M56	335	Ø122	90.6	112.6	8	20	Ø116	6.14	7.7
EZI-CI204U	2.25	335	Ø122	90.6	112.6	8	20	Ø116	6.13	7.66
EZI-CI208U	2.5	439	Ø136	96.1	116.1	8	20	Ø132	8.17	9.88
EZI-CM64X6	M64	439	Ø136	96.1	116.1	8	20	Ø132	8.13	9.86
EZI-CI212U	2.75	527	Ø150	105	122.9	8	20	Ø146	10.86	12.83
EZI-CM72X6	M72	527	Ø150	105	122.9	8	20	Ø146	10.74	12.68
EZI-CI300U	3	638	Ø160	107	124	8	20	Ø156	12.57	14.66
EZI-CM80X6	M80	706	Ø170	108.5	125.5	8	20	Ø162	14.38	16.77

Commercial grade Imperial EziTite® nuts are calculated at approximately 65% proof load of a SAE Grade 2 bolt. Commercial grade Metric EziTite® nuts are calculated at approximately 65% proof load of a Grade 4.6 bolt. **Maximum operating pressure for commercial grade is 700 Bar**

HIGH TENSILE GRADE : Max. operating pressure 1,000 Bar

Model Part #	Thread Size	Force (kN)	(A) Nut OD (mm)	(B) Height Short Stroke (mm)	(B) Height Long Stroke (mm)	Short Stroke (mm)	Long Stroke (mm)	(C) Washer Diameter	Weight Approx. Short (kg)	Weight Approx. Long (kg)
EZI-HM24X3	M24	160	Ø70	59.9	72	6	12	Ø66	1.52	1.84
EZI-HI100U	1	160	Ø70	59.9	72	6	12	Ø66	1.52	1.82
EZI-HI102U	1.125	171.9	Ø74	62.3	74.3	6	12	Ø68	1.67	2.01
EZI-HM30X35	M30	249.3	Ø84	65	77	6	12	Ø80	2.32	2.79
EZI-HI104U	1.25	249.3	Ø84	65	77	6	12	Ø80	2.3	3
EZI-HI106U	1.375	277.9	Ø92	68.3	80.3	6	12	Ø88	2.89	3.42
EZI-HM36X4	M36	357	Ø100	69.6	80.6	6	12	Ø96	3.52	4.09
EZI-HI108U	1.5	332.5	Ø100	69.6	80.6	6	12	Ø96	3.46	4.03
EZI-HI110U	1.625	409	Ø110	78.9	88.9	6	12	Ø106	4.68	5.34
EZI-HM42X45	M42	456	Ø114	78.3	88.3	6	12	Ø110	5.02	5.74
EZI-HI112U	1.75	456	Ø114	80.3	103.3	8	20	Ø110	5.11	6.66
EZI-HM48X5	M48	629.3	Ø128	84	106	8	20	Ø124	6.84	8.72
EZI-HI200U	2	598.5	Ø128	84	106	8	20	Ø124	6.74	8.61
EZI-HM56X55	M56	821.1	Ø146	89.5	110.5	8	20	Ø142	9.52	11.9
EZI-HI204U	2.25	785.4	Ø146	89.5	110.5	8	20	Ø142	9.47	11.81
EZI-HI208U	2.5	940.5	Ø156	95.6	115.6	8	20	Ø152	11.47	13.9
EZI-HM64X6	M64	1108.4	Ø166	95.8	115.8	8	20	Ø162	13.26	16.05
EZI-HI212U	2.75	1128.3	Ø172	100.6	120.6	8	20	Ø168	14.82	17.85
EZI-HM72X6	M72	1474.9	Ø190	99.5	119.5	8	20	Ø186	18.29	22.09
EZI-HI300U	3	1379.9	Ø190	99.9	119.9	8	20	Ø186	17.99	21.78
EZI-HM80X6	M80	1770.3	Ø206	105.9	124.9	8	20	Ø202	22.8	26.98

High Tensile Grade Imperial are calculated at approximately 65% proof load of a SAE Grade 5 bolt. High Tensile Grade Metric are calculated at approximately 65% proof load of a Grade 8.8 bolt. **Maximum operating pressure for High Tensile Grade is 1000 Bar.** If the standard range above does not suit, our technical staff will modify the design to suit the application. Long Stroke part numbers have an 'L' at the end of the part number. e.g. EZI-CI112UL.

All standard EziTite® Hydraulic Nuts are equipped with CEJN type male snap fittings, 1/8" BSPP porting and bleed plugs:

- **EziTite® Hydraulic Nuts** are supplied with spherical self-aligning washers as standard.
- Maximum force is generated using the stated maximum pressure.

EziTite® TR High Temperature Hydraulic Nut

- High-Temperature Resistant (to 550°C+)
- Secondary Backup Release Mechanism
- Patented Metal-to-Metal Seal Design
- Eliminate Load Losses

EZITITE®

The patented design of the high-temperature resistant **EziTite® TR High Temperature Hydraulic Nut** has given it improved technical efficiency of operation and cost/benefit advantages over other tensioning devices.

Operation

The **EziTite® TR High Temperature Hydraulic Nut** operates by providing simultaneous multiple tensioning to ensure even bolt loads, which significantly streamlines assembly and reduces disassembly downtime.

Its smart design is specifically engineered to eliminate load losses during the tensioning process, providing a reliable and precise fit with minimal physical effort.

In the event of service damage or misalignment which may disrupt seal integrity, the unit features a unique Sacrificial Ring.

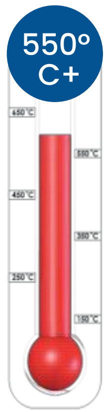
By carefully splitting and removing this ring, the **EziTite® TR High Temperature Hydraulic Nut** will retract normally for safe removal.

This efficient operational cycle makes it the ideal solution for high-pressure and high-temperature applications such as refineries, processing and steelworks, pressure vessel closures (i.e. Autoclaves, Boilers, etc), flange makeup, steam turbine half-joints and gas turbine casings.

Features

The **EziTite® TR High Temperature Hydraulic Nut** is defined by its patented metal-to-metal seal design, which offers superior technical efficiency and reliable performance in extreme environments up to 550°C, including:

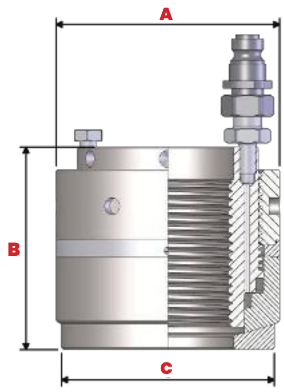
- Patented Metal-to-Metal Seals
- High-Temperature Resistance (to 550°C and higher)
- Compact Design for Confined Spaces
- Smart Design to Eliminate Load Losses
- Secondary Backup Release Mechanism (Sacrificial Ring)
- Rapid Assembly and Disassembly Capability
- User-Friendly and Safe Operation



Specifications

Model Part #	Thread Size	Force (kN)	(A) Nut OD (mm)	(B) Height (mm)	Stroke (mm)	(C) Washer Diameter	Weight (kg)
H-EZI-M24X25	M24	380.1	Ø74.0	87.2	8	Ø71.0	2.41
H-EZI-I100X8U	1	380.1	Ø74.0	87.2	8	Ø71.0	2.41
H-EZI-I102X8U	1.125	390	Ø77.0	87	8	Ø74.0	2.56
H-EZI-M30X35	M30	400	Ø80.0	87.9	8	Ø77.0	2.72
H-EZI-I104X8U	1.25	400	Ø80.0	87.9	8	Ø77.0	2.72
H-EZI-I106X8U	1.375	410	Ø83.0	88.7	8	Ø80.0	2.91
H-EZI-M36X40	M36	455.9	Ø90.0	89.2	8	Ø87.0	3.4
H-EZI-I108X8U	1.5	455.9	Ø90.0	89.2	8	Ø87.0	3.4
H-EZI-I110X8U	1.625	542	Ø97.0	90.6	8	Ø94.0	3.97
H-EZI-M42X45	M42	635.6	Ø103.0	91.1	8	Ø100.0	4.52
H-EZI-I112X8U	1.75	635.6	Ø103.0	91.1	8	Ø100.0	4.52
H-EZI-M48X50	M48	845.2	Ø117.0	91.9	8	Ø114.0	5.88
H-EZI-I200X8U	2	845.2	Ø117.0	91.9	8	Ø114.0	5.88
H-EZI-M56X55	M56	1084.6	Ø132.0	96.5	8	Ø129.0	7.77
H-EZI-I204X8U	2.25	1084.6	Ø132.0	96.5	10	Ø129.0	7.77
H-EZI-I208X8U	2.5	1353.7	Ø146.0	103.4	10	Ø143.0	10.26
H-EZI-M64X60	M64	1353.7	Ø146.0	103.4	10	Ø143.0	10.26
H-EZI-I212X8U	2.75	1495.3	Ø148.0	105.7	10	Ø145.0	10.55
H-EZI-M72X60	M72	1792.8	Ø163.0	108.4	10	Ø160.0	10.55
H-EZI-I300X8U	3	1792.8	Ø163.0	108.4	10	Ø160.0	13.14
H-EZI-M80X60	M80	2044.2	Ø172.0	110.3	10	Ø169.0	15.83
H-EZI-I308X8U	3.5	2468.6	Ø188.0	113.3	10	Ø185.0	18.26

If the standard nut does not fit within your space restrictions or load requirements other models and custom designs are available upon request.



Secondary release mechanism

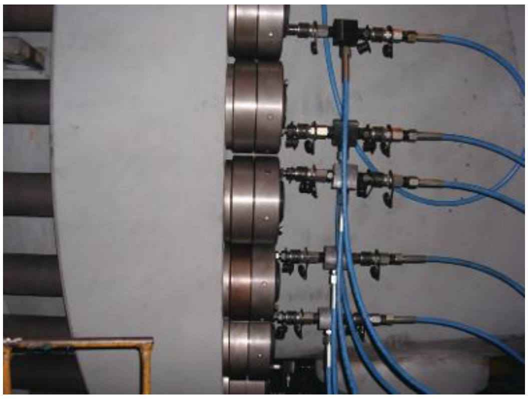
If the **EziTite® TR High Temperature Hydraulic Nut** is damaged in service (eg. by a misalignment that compromises seal integrity), it can be removed by carefully splitting the Sacrificial Ring.

Once this ring is removed, the unit will retract normally for safe removal.

1. Isolate the nut

2. Break the sacrificial ring

3. The nut will now release the load



EziTite® Bearing Setter

- Eliminate Premature Bearing Failure
- Precision Hydraulic Mounting & Preload
- Unique Split Design for Rapid Resets
- Effortless Operation in Confined Spaces

EZITITE®

Major bearing manufacturers estimate that 16% of bearings fail prematurely due to poor fitting, leading to expensive breakdowns and rework.

The **EziTite® Bearing Setter** uses hydraulic power to guide bearings into place effortlessly, providing the optimum preload required for reliable operation.

Designed for the operator, this tool ensures accurate, repeatable mounting with minimal physical effort, even in difficult or confined spaces, and offers various optional extras to ensure trouble-free performance.

Operation

The **EziTite® Bearing Setter** assembly is screwed by hand onto the shaft or adapter sleeve *(Fig. 1a)* until the base is tight against the bearing's inner race. *(Fig. 1b)*

Hydraulic pressure is then applied through the nipple fitting (located on the top or side of the nut body) into the sealed chamber forcing the piston and the nut body apart, and driving the bearing into position *(Fig. 2)*.

Once the desired bearing drive-up has been achieved, the pressure is relieved, and the Bearing Setter is removed and replaced with a standard locking nut *(Fig. 3)*.

For added efficiency, Technofast's unique split design can be placed over the shaft for bearing preload resets without the need to disassemble the entire machine.

Features

The EziTite® Bearing Setter has been designed with the operator in mind, offering many optional extras readily available to ensure trouble-free operation.

Some features of the **EziTite® Bearing Setter** are:

- Standard Dial Gauge Mounting Point
- Visible Maximum Stroke Indicator
- Custom Thread Sizes Available
- Optional Split Nut Designs
- Engineered Customisation Available
- Minimal Physical Effort Required

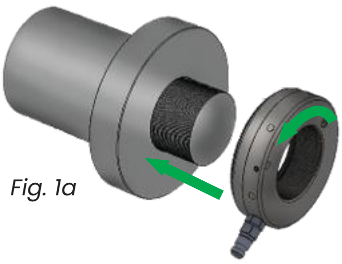


Fig. 1a

Fig. 1b

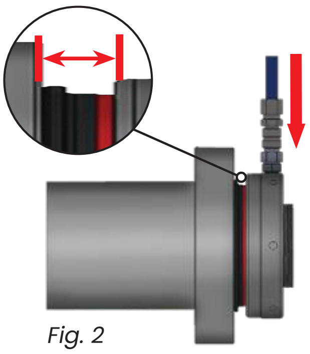
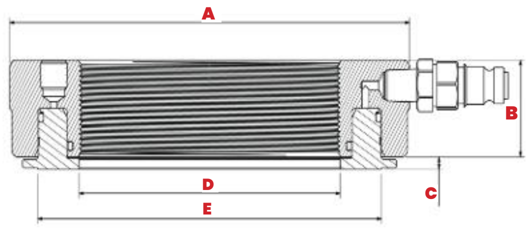


Fig. 2

Fig. 3



Specifications

Tool # (Metric)	Size ISO	Thread Pitch mm	Tool # Imperial	Size Imp	Threads Per Inch	Max Force kN	HPA	Pressure (Bar)	(A)	(B)	(C)	(D)	(E)	Pin Spanner Hole Size	Stroke
BP-03-M00	M60	2	BP-03-U00	2.36	18	230.6	3293.6	700	120	38	4	60.5	96	8mm	7mm
BP-04-M00	M65	2	BP-04-U00	2.548	18	248.3	3546.95	700	126	38	4	65.5	102	8mm	7mm
BP-05-M00	M70	2	BP-05-U00	2.751	18	257.2	3673.83	700	132	38	4	70.5	108	8mm	7mm
BP-06-M00	M75	2	BP-06-U00	2.933	12	274.9	3927.18	700	136	38	4	75.5	116	8mm	7mm
BP-07-M00	M80	2	BP-07-U00	3.137	12	292.7	4180.54	700	142	38	5	80.5	118	8mm	7mm
BP-08-M00	M85	2	BP-08-U00	3.34	12	310.4	4433.89	700	148	38	5	85.5	124	8mm	7mm
BP-09-M00	M90	2	BP-09-U00	3.527	12	319.2	4560.77	700	154	38	5	90.5	132	8mm	7mm
BP-10-M00	M95	2	BP-10-U00	3.73	12	337	4813.72	700	160	38	5	95.5	138	8mm	7mm
BP-11-M00	M100	2	BP-11-U00	3.918	12	355.8	5075.85	700	164	44	6	100.5	140	8mm	7mm
BP-12-M00	M105	2	BP-12-U00	4.122	12	327.4	5456.7	600	176	45	5	105.5	154	8mm	7mm
BP-13-M00	M110	2	BP-13-U00	4.325	12	327.4	5456.7	600	176	45	5	110.5	154	8mm	8mm
BP-14-M00	M115	2	BP-14-U00			365.5	6090.86	600	190	45	5	115.5	166	8mm	8mm
BP-15-M00	M120	2	BP-15-U00	4.716	12	365.5	6090.86	600	190	45	5	120.5	166	8mm	8mm
BP-16-M00	M125	2	BP-16-U00			364.8	6080.49	600	195	45	5	125.5	172	8mm	8mm
BP-17-M00	M130	2	BP-17-U00	5.106	12	423.2	7053.2	600	200	45	5	130.5	172	8mm	8mm
BP-18-M00	M135	2	BP-18-U00			411.1	6852.12	600	208	45	5	135.5	186	8mm	8mm
BP-19-M00	M140	2	BP-19-U00	5.497	12	411.1	6852.12	600	208	45	5	140.5	186	8mm	8mm
BP-20-M00	M145	2	BP-20-U00			426.4	7105.88	600	222	46	6	145.5	199	8mm	8mm
BP-21-M00	M150	2	BP-21-U00	5.888	12	426.4	7105.88	600	222	46	6	150.5	199	8mm	8mm
BP-22-M00	M155	3	BP-22-U00			580.3	9672.28	600	234	47	6	155.5	210	8mm	8mm
BP-23-M00	M160	3	BP-23-U00	6.284	8	580.3	9672.28	600	234	47	6	160.5	210	8mm	8mm
BP-24-M00	M165	3	BP-24-U00			618.4	10306.46	600	250	47	6	165.5	224	8mm	8mm
BP-25-M00	M170	3	BP-25-U00	6.659	8	618.4	10306.46	600	250	47	6	170.5	224	8mm	8mm
BP-26-M00	M175	3	BP-26-U00			637.4	10623.55	600	256	50	6	175.5	230	8mm	8mm
BP-27-M00	M180	3	BP-27-U00	7.066	8	637.4	10623.55	600	256	50	6	180.5	230	10mm	10mm
BP-28-M00	M185	3	BP-28-U00			675.5	11257.74	600	268	50	6	185.5	242	10mm	10mm
BP-29-M00	M190	3	BP-29-U00	7.472	8	675.5	11257.74	600	268	50	6	190.5	242	10mm	10mm
BP-30-M00	M195	3	BP-30-U00			713.5	11891.92	600	282	51	7	195.5	256	10mm	10mm
BP-31-M00	M200	3	BP-31-U00	7.847	8	713.5	11891.92	600	282	51	7	200.5	256	10mm	10mm
BP-32-M01	TR205	4	BP-32-U00			731.6	12192.65	600	290	51	7	205.5	262	10mm	10mm
BP-33-M01	TR210	4	BP-33-U00			731.6	12192.65	600	290	51	7	210.5	262	10mm	10mm
BP-34-M01	TR215	4	BP-34-U00			770.6	12843.19	600	304	53	9	215.5	274	10mm	10mm
BP-35-M01	TR220	4	BP-35-U00	8.628	8	770.6	12843.19	600	304	53	9	220.5	274	10mm	10mm
BP-36-M01	TR225	4	BP-36-U00			807.6	13459.42	600	315	53	9	225.5	286	10mm	10mm
BP-37-M01	TR230	4	BP-37-U00			807.6	13459.42	600	315	53	9	230.5	286	10mm	10mm
BP-38-M01	TR235	4	BP-38-U00			846.7	14111.56	600	334	55	9	235.5	300	10mm	10mm
BP-39-M01	TR240	4	BP-39-U00	9.442	6	846.7	14111.56	600	334	55	9	240.5	300	10mm	10mm
BP-40-M01	TR250	4	BP-40-U00			956.7	14726.19	600	359	54	9	250.5	325	12mm	12mm
BP-41-M01	TR260	4	BP-41-U00	10.192	6	956.7	15945.62	600	359	54	9	260.5	325	12mm	12mm
BP-42-M01	TR270	4	BP-42-U00	10.604	6	960.9	16014.1	600	372	58	9	272	338	12mm	12mm
BP-43-M01	TR280	4	BP-43-U00	11.004	6	960.9	16014.1	600	372	58	9	282	338	12mm	12mm
BP-44-M01	TR290	4	BP-44-U00			1140.1	19001.11	600	404	59	9	292	365	12mm	14mm
BP-45-M01	TR300	4	BP-45-U00	11.785	6	1140.1	19001.11	600	404	59	9	302	365	12mm	14mm
BP-46-M01	TR310	5	BP-46-U00			1238.8	20647.89	600	412	50	9	312	374	12mm	14mm
BP-47-M01	TR320	5	BP-47-U00	12.562	6	1463	24383.48	600	430	52	10	322	390	12mm	14mm
BP-48-M01	TR330	5	BP-48-U00			1508.6	25142.89	600	440	52	10	332	400	12mm	14mm
BP-49-M01	TR340	5	BP-49-U00	13.339	5	1578.4	26307.37	600	454	52	10	342	412	12mm	14mm
BP-50-M01	TR350	5	BP-50-U00			1594.3	26571.84	600	466	52	10	352	424	12mm	14mm
BP-51-M01	TR360	5	BP-51-U00	14.17	5	1614.1	26902.33	600	472	52	10	362	430	12mm	14mm
BP-52-M01	TR370	5	BP-52-U00			1635.9	27266.12	600	490	56	12	372	448	12mm	16mm
BP-53-M01	TR380	5	BP-53-U00	14.957	5	1709.3	28488.71	600	500	56	12	382	454	12mm	16mm
BP-54-M01	TR390	5	BP-54-U00			1776.9	29616.1	600	512	56	12	392	464	12mm	16mm
BP-55-M01	TR400	5	BP-55-U00	15.745	5	1804.7	30078.56	600	522	56	12	402	476	12mm	16mm

If the standard range above does not suit, our technical staff will modify the design to suit the application. For specialised thread and sizing requirements, don't hesitate to contact sales@technofast.com for assistance.

EziTite® Hydraulic Bolt

- Vibration & Shock Resistant
- Unique Load-Retaining Lockring
- Galling and Torsion Resistant
- Quick-Connect Compatibility

EZITITE®

The **EziTite® Hydraulic Bolt** is a precision-engineered, high-performance fastener incorporating a hydraulically operated tensioning mechanism Designed to replace standard fasteners, these bolts are extremely resistant to vibration and shock loads, making them suitable for vibrating screens, sieves, and foundation bolts.

They are manufactured in a choice of alloy or stainless steel with tensile strengths to suit the required application and can be quickly fitted using standard pumping equipment, such as Technofast's hand-operated, electric, or air-hydraulic pumps.

Operation

The **EziTite® Hydraulic Bolt** replaces the standard fastener in an application.

Hydraulic pressure is applied through the quick-connect fitting into the sealed chamber, forcing the nut body and piston apart to tension the bolt through the joint.

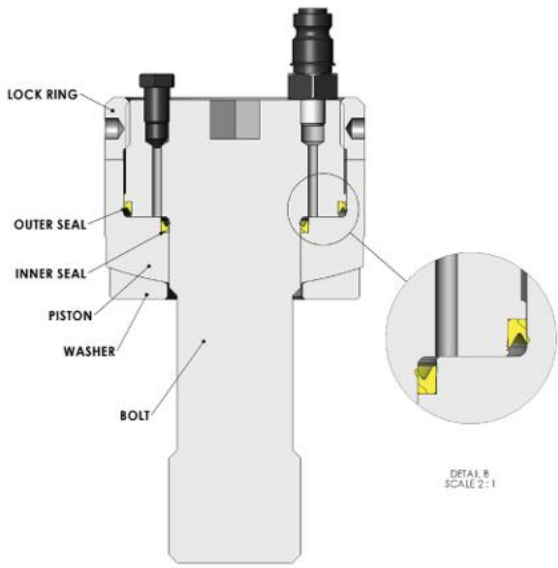
To retain the induced load, the threaded lockring is screwed against the abutting face. Once pressure is released and the coupling disconnected, the operation is complete.

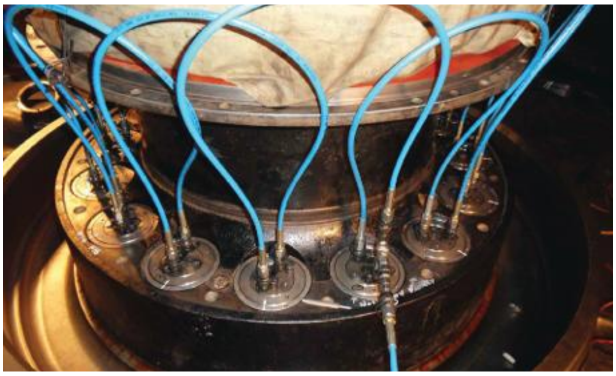
This method ensures accurate loading with no thread galling or torsional stress. It also allows for easy removal, even in confined or difficult locations as the lockring unscrews without friction by simply reversing the process.

Features

The **EziTite® Hydraulic Bolt** has been designed with the operator in mind, offering many optional extras readily available to ensure trouble-free operation. Some features of the **EziTite® Hydraulic Bolt** are:

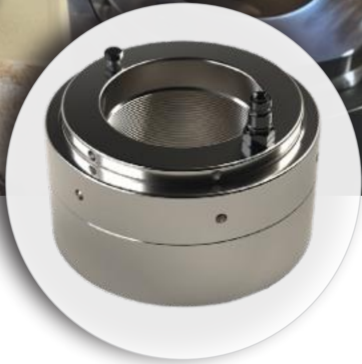
- Standard range covers 20mm to 100mm diameters
- Unique Lockring ensures maximum retained load
- Spherical seats provide alignment to joint faces
- Various seal designs meet specific temperature requirements
- Quick-connect fittings and bleed plugs are standard
- Stroke indicators are fitted to all standard bolts
- Max. pressure reaches 100 MPa for standard designs
- Versatile design suits piping flanges and reactor joints





EziTite® Hydraulic Clamp Nut

- **Precise and Repeatable Tool Clamping**
- **Eliminates Conventional Torque Loading**
- **Impact-Resistant Threaded Locking Collar**



The **EziTite® Hydraulic Clamp Nut** is a modern, safe, and efficient solution for ensuring precise tool clamping.

Designed to replace conventional torque-loaded threaded nuts, these units make shaft tensioning a simple and safe hydraulically assisted operation.

They are widely used to set large shaft bearings and are manufactured in a standard range to suit thread sizes from 50mm to 300mm, with other sizes available on request.

Operation

The **EziTite® Hydraulic Clamp Nut** is screwed by hand onto the bolt until the base is tight against the working face.

Hydraulic pressure is applied through the nipple fitting into the sealed chamber, forcing the piston and nut body apart to apply a specified joint clamping force.

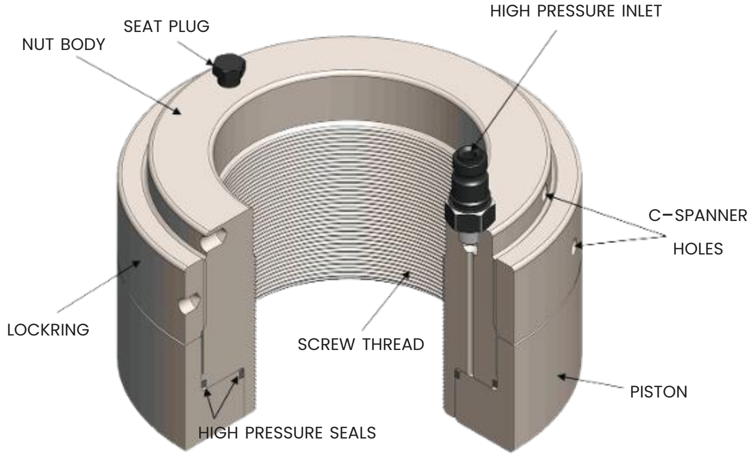
To retain the induced load, the threaded lockring on the nut body is screwed against the abutting face.

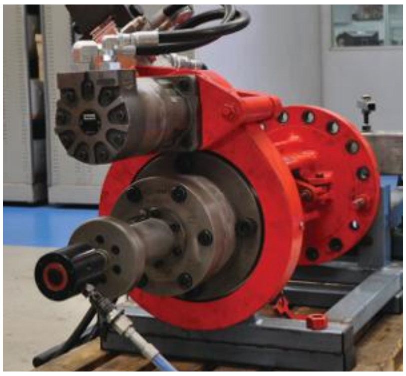
Once pressure is released and the coupling is removed, the operation is complete.

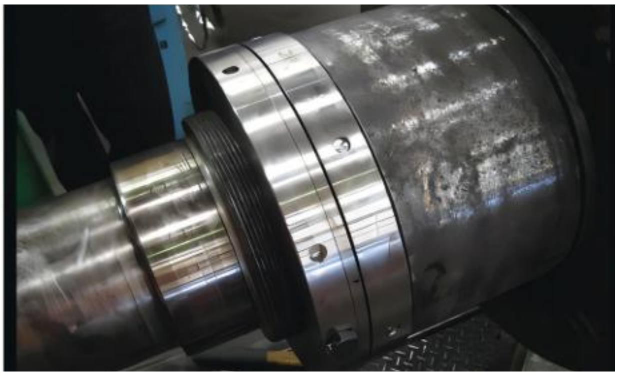
Features

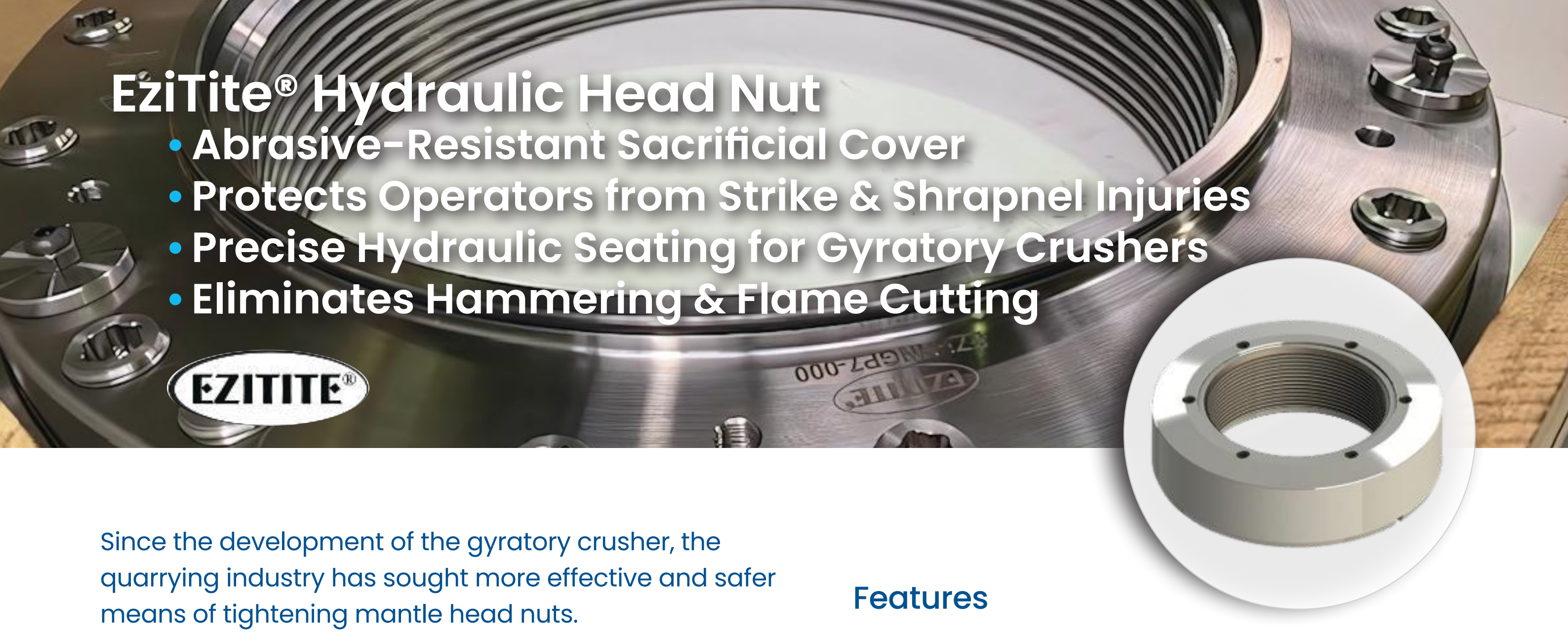
The **EziTite® Hydraulic Clamp Nut** has been designed with the operator in mind, offering many optional extras readily available to ensure trouble-free operation. Some features of the **EziTite® Hydraulic Clamp Nut** are:

- Standard range fits 50mm to 300mm threads
- Precision clamping increases tool life
- Repeatable accuracy reduces product variation
- Fast fitting achieves high throughput
- User-friendly design prioritises operator safety
- Lockring counteracts rotating impact forces
- Hydraulic operation reduces maintenance downtime









EziTite® Hydraulic Head Nut

- Abrasive-Resistant Sacrificial Cover
- Protects Operators from Strike & Shrapnel Injuries
- Precise Hydraulic Seating for Gyratory Crushers
- Eliminates Hammering & Flame Cutting

EZITITE®

Since the development of the gyratory crusher, the quarrying industry has sought more effective and safer means of tightening mantle head nuts.

The **EziTite® Hydraulic Head Nut** system was successfully developed by Technofast to replace standard mantle head nuts.

A primary consideration of this design is the protection of operators from strike or shrapnel injuries commonly associated with hammer tightening.

Operation

The **EziTite® Hydraulic Head Nut** assembly is screwed by hand onto the crusher shaft until the base is tight against the mantle's working face.

Hydraulic pressure is applied through a nipple fitting into the sealed chamber, forcing the piston and nut body apart to seat the mantle onto the machine's taper.

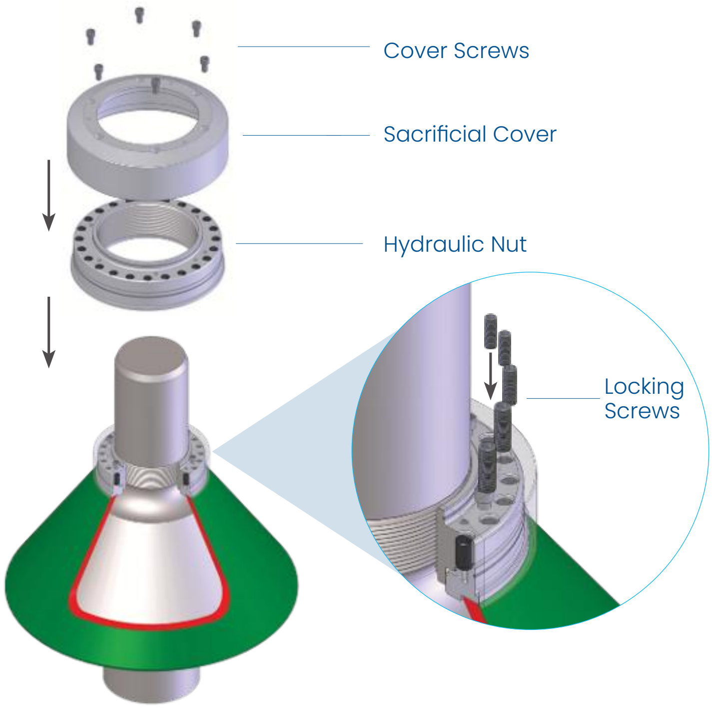
Mechanical locking screws or rings are then engaged to retain the tensile loads. To remove the assembly, the process is reversed and the nut is removed by hand.

This eliminates the need for large hammers or flame cutting of the burn ring, significantly improving site safety and reducing maintenance downtime.

Features

The **EziTite® Hydraulic Head Nut** is designed for high-performance clamping in extreme quarrying environments. Some features include:

- Mechanical locking screws or rings retain all tensile loads
- Threaded collar counteracts rotating impact forces
- Simple hand-operated installation
- Customisable for material abrasiveness



Patent Protection: Australian and International Patents Granted & Pending.
Patent No: PCT/WO2006/037173


3M

CamNut™

- Eliminates Thread Galling and Increases Accuracy
- Extreme Temp Resistance (-40°C to +650°C)
- Modular System for Simultaneous Loading
- Enables Tensioning on Short Bolt Grips

Technofast's **CamNut™** tensioning system is a revolutionary, compact solution designed to provide affordable and efficient bolt tensioning without the need for hardware modifications.

Capable of operating in extreme temperatures ranging from -40°C to +650°C, the H1 and H2 series fit standard OEM machine specifications without altering existing bolts or spot faces.

This system allows users to apply direct hydraulic tensioning to turbines, valves, and other critical hardware, eliminating thread galling and increasing bolt load accuracy.

Operation

Standard bolt tensioning requires additional thread length above a hex nut for a tool to grip. The **CamNut™** removes this requirement by replacing the standard nut and serving as the direct attachment point for a separate hydraulic bolt tensioner.

During operation, the external tensioner pulls on the bolt through the **CamNut™**, and the nut's integral collar expands to take up the bolt's elongation.

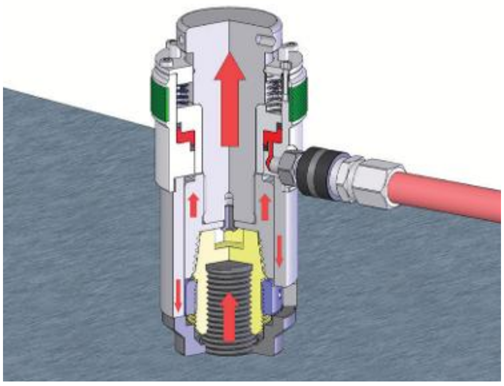
Once the hydraulic pressure is released and the load is retained by the collar, the tensioner is removed for use on the next bolt. This modular system allows for the simultaneous tensioning of multiple bolts, ensuring even joint and gasket compression.

Features

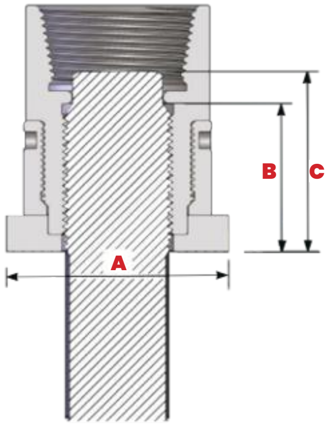
The patented **CamNut™** system is designed for ease of use in restricted spaces or adverse conditions.

Some features of the **CamNut™** are:

- Modular construction reduces overall tool weight
- Quick nut-to-tensioner connection saves time
- Ideal for short bolt grip lengths
- Operates from -40°C up to +650°C
- Fast fitting requires little physical effort
- Eliminates expensive stud replacement for grip
- Fits standard OEM machines without modifications
- Suits large bolts and simultaneous tightening modifications
- Suits large diameter bolts and simultaneous tightening

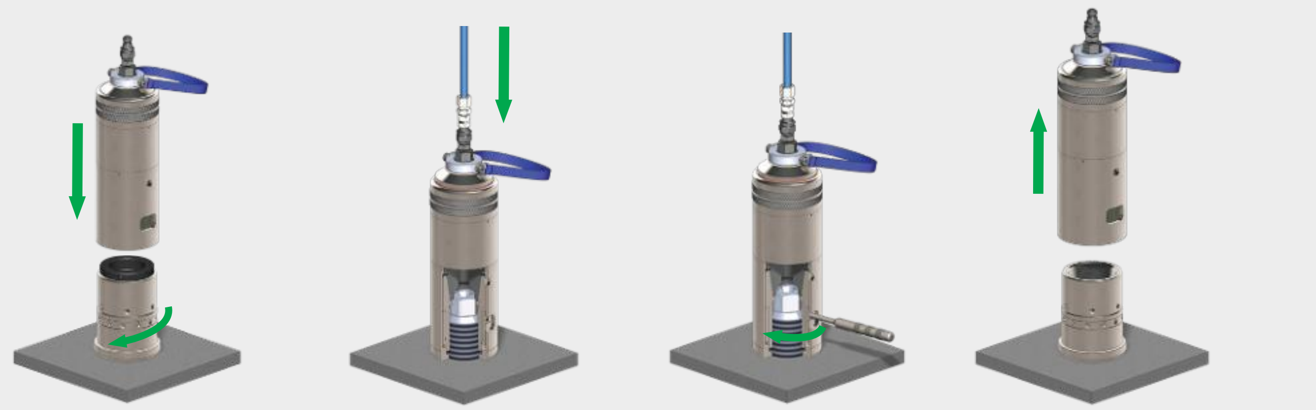


Specifications



Tensioner	Model Part # -40°C to 400°C	Model Part # 400°C to 560°C	Thread Size	Force Produced (kN)	(A) Nut OD (mm)	(B) Max Stud Length	(C) Max Stud Length with Hex	Nut Weight (kg)
CT-HS-04	H1-CN-112X8	H2-CN-112X8	1.75"	587.03	Ø74	64.0	66.55	2.03
CT-HS-05	H1-CN-200X8	H2-CN-200X8	2"	814.14	Ø84	70.0	73.15	2.75
CT-HS-06	H1-CN-204X8	H2-CN-204X8	2.25"	1011.14	Ø94	72.0	82.55	3.44
CT-HS-07	H1-CN-208X8	H2-CN-208X8	2.5"	1245.75	Ø104	79.0	88.9	4.47
CT-HS-08	H1-CN-212X8	H2-CN-212X8	2.75"	1418.95	Ø114	85.0	98.55	5.60
CT-HS-09	H1-CN-300X8	H2-CN-300X8	3"	1788.20	Ø124	86.0	98.55	7.07
CT-HS-10	H1-CN-308X8	H2-CN-308X8	3.5"	2425.10	Ø144	94.0	117.35	9.90
CT-HS-11	H1-CN-400X8	H2-CN-400X8	4"	3041.60	Ø164	107.0	130.05	14.45
CT-HS-12	H1-CN-408X8	H2-CN-408X8	4.5"	4069.32	Ø183	119.0	146.05	18.79
CT-HS-13	H1-CN-500X8	H2-CN-500X8	5"	4782.12	Ø203	129.0	158.75	25.13
CT-HS-14	H1-CN-508X8	H2-CN-508X8	5.5"	6112.58	Ø223	145.0	177.80	33.72
CT-HS-15	H1-CN-600X8	H2-CN-600X8	6"	7088.87	Ø244	154.0	190.50	41.53

If the standard **CamNut™** does not meet your space or load requirements, custom designs are available on request. For detailed technical specifications and further information, please contact Technofast Industries. This product is protected by Worldwide Patents. **Patent No: PCT/WO2005/123345**

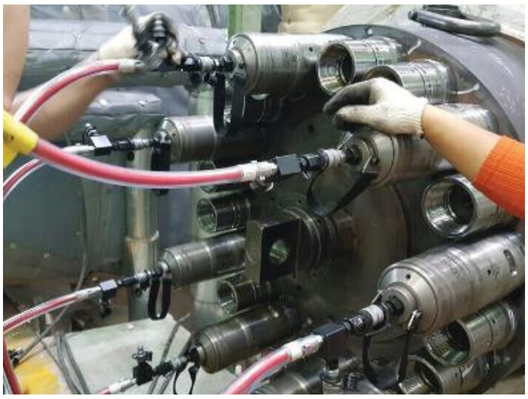


1. Fasten **CamNut™** onto bolt stud before fastening tensioner tool into place
2. Connect hose and pressurise tensioner
3. Screw lockring against washer
4. Release pressure and disconnect hose to remove tensioner tool

H1 & H2 Series Comparison

The H1 and H2 series are engineered to fit within standard OEM machine specifications without hardware modifications:

Series	Temperature Range	Application Focus
H1 Series	-40°C to +560°C	Standard turbines, valves, and OEM hardware
H2 Series	-40°C up to +650°C	High-temperature and extreme environment hardware



EziJac B2 Dedicated Hydraulic Bolt Tensioner

- Interlink for Simultaneous Tensioning
- Economical Dedicated Stud Solution
- 150 MPa Maximum Force Capacity
- Leak-Free Low-Friction Seal Design

The **B2 EziJac Bolt Tensioner** is a dedicated series of tools, where each unit is engineered to provide an economical and precise tensioning solution for a specific bolt size and thread type.

Utilising hydraulic pressure to tension studs accurately and safely, the **B2** Series features a manual return system and a compact design to resolve fitting challenges often associated with larger equipment.

These tools are ideal for single-task applications where thread interchangeability is not required, and any number of units can be interlinked to provide simultaneous tensioning across multiple bolts.

Operation

The **B2 EziJac** produces the necessary force to tension bolts without inducing rotational or torsional stresses. This manual return tensioner is designed for accuracy and repeatability, making it an ideal replacement for traditional hammer tightening or bolt heating methods.

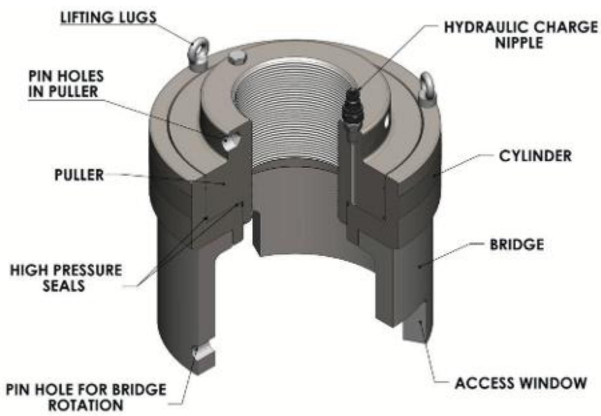
For complex applications like flange makeup, multiple **B2 EziJac** units can be connected to ensure simultaneous tensioning and even joint loading.

The tool is particularly effective for regular maintenance cycles requiring repeated adjustments, providing reliable and precise tensioning even in confined or difficult nut locations.

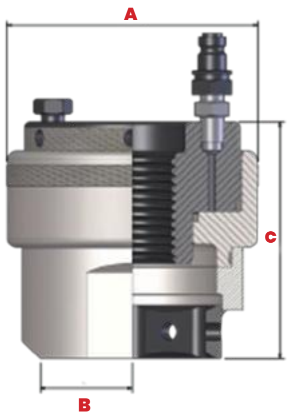
Features

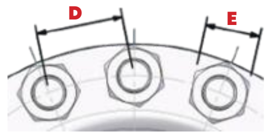
The **B2 EziJac** is built for a long service life and user-friendly operation in demanding environments. Some features of the **B2 EziJac** are:

- Dedicated design ensures an economical tool for specific bolt sizes
- Long 15mm stroke provided across all models
- Integrated stroke indicator for easy monitoring
- Leak-free low-friction seals for reliable performance
- Nitrile components increase overall tool life and durability
- Optional second port allows for multi-tool interconnection
- Maximum force generated at 150 MPa (1500 Bar)
- Compact profile suits confined spaces and difficult locations

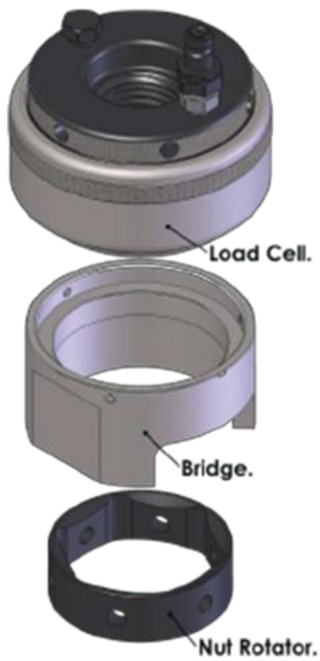


Specifications





Model #	Size Imp	Size Metric	Max Force (kN)	(A) O.D.	(D) Min Pitch	(B) Relief	(C) Height	(E) Nut A.F.	Weight (kg)
B2-I014X9	7/8"	22.2	285.7	72	53	26	89	36.5	1.78
B2-M24X30		M24	285.7	72	57	29	90	41	1.78
B2-I100X8	1"	25.4	367.4	79	57	29	102	41.3	2.48
B2-M27X30		M27	367.4	79	64	32	103	46	2.48
B2-I102X8	1.1/8"	28.5	428.1	86	64	32	105	46	2.96
B2-M30X35		M30	428.1	86	69	36	107	50	2.96
B2-I104X8	1.1/4"	31.75	570.3	97	70	36	111	50.8	3.94
B2-M33X35		M33	570.3	97	72	36	113	55	3.94
B2-I106X8	1.3/8"	34.9	664.6	104	74	39	114	55.6	4.52
B2-M36X40		M36	664.6	104	77	39	116	60	4.52
B2-I108X8	1.1/2"	38.1	765.7	112	79	42	120	60.3	5.39
B2-M39X40		M39	765.7	112	82	42	120	65	5.39
B2-M42X45		M42	909.6	122	91	45	125	70	6.5
B2-I110X8	1.5/8"	41.27	909.6	122	89	45	124	65.1	6.5
B2-M45X45		M45	1027.2	129	91	49	128	75	6.7
B2-I112X8	1.3/4"	44.45	1027.2	129	94	49	128	69.9	7.56
B2-M48X50		M48	1192.2	139	103	51	132	80	8.93
B2-I114X8	1.7/8"	47.6	1192.2	139	100	51	132	74.6	8.93
B2-M52X50		M52	1283	143	106	54	136	85	9.41
B2-I200X8	2"	50.8	1283	143	103	54	135	79.4	9.41
B2-M56X55		M56	1456.6	158	117	60	142	90	12.13
B2-I204X8	2.1/4"	57.15	1456.6	158	116	60	144	88.9	12.13
B2-M60X55		M60	1456.6	158	126	65	146	95	12.13
B2-M64X60		M64	1822.2	173	129	65	152	100	15.16
B2-I208X8	2.1/2"	63.5	1822.2	173	128	65	152	98.4	15.16
B2-M68X60		M68	2213.9	192	142	71	158	105	19.53
B2-M72X60		M72	2213.9	192	145	71	162	110	19.53
B2-I212X8	2.3/4"	69.85	2213.9	192	143	71	161	108	19.53
B2-I300X8	3"	76.2	2657.1	206	155	77	171	117.5	23.24
B2-M76X60		M76	2657.1	206	153	77	170	115	23.24
B2-M80X60		M80	2657.1	206	170	77	174	120	23.24
B2-I304X8	3.1/4"	82.55	3061	220	170	87	181	127	28.14
B2-M85X60		M85	3061	220	169	87	182	125	28.14
B2-I308X8	3.1/2"	88.9	3720.2	242	175	93	190	136.5	34.63
B2-M90X60		M90	3720.2	242	174	93	191	135	34.63
B2-I312X8	3.3/4"	95.25	4195.3	256	196	99	200	146.1	40.93
B2-M95X60		M95	4195.3	256	196	99	200	145	40.93
B2-I400X8	4"	101.6	4605.3	269	209	105	207	155.6	47.61
B2-M100X60		M100	4605.3	269	206	105	205	150	47.61

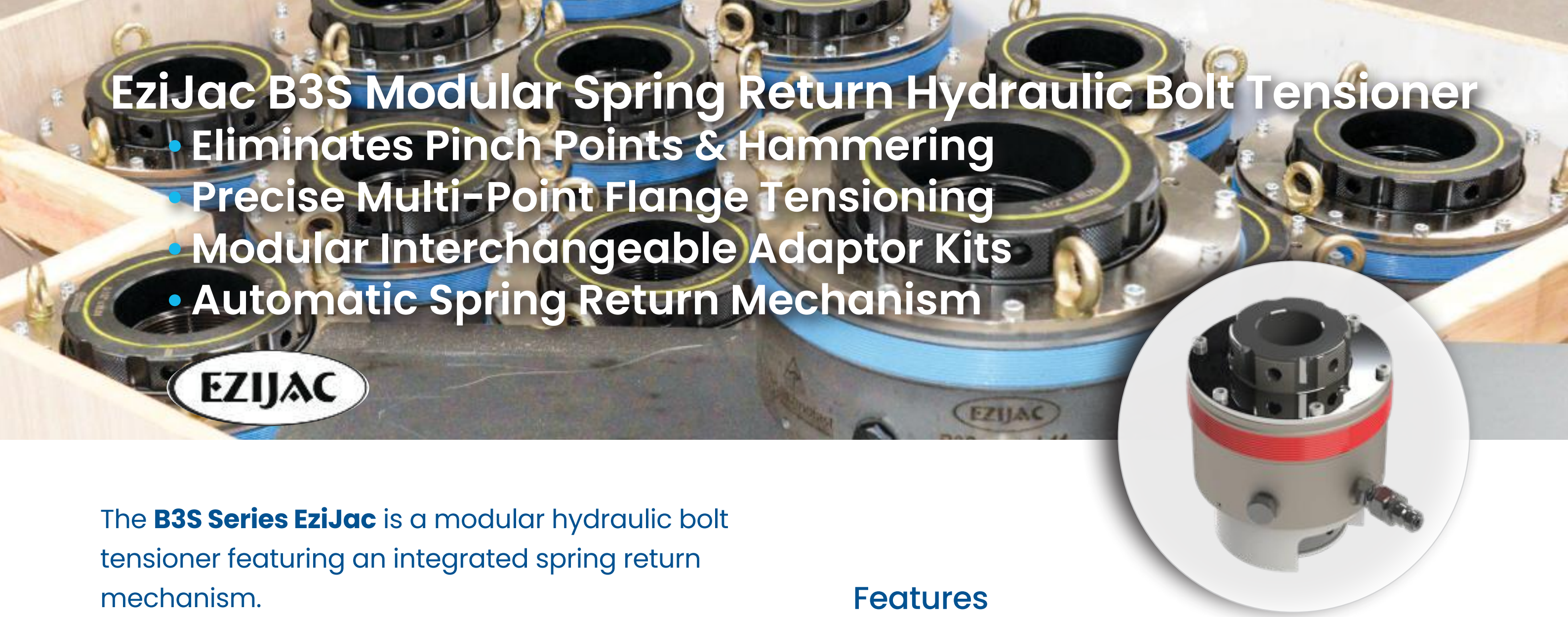


Stroke for all Models is 15.0 mm. Maximum Force is generated at 150 MPa (1500 Bar)
Dimensions in millimetres unless noted otherwise. These dimensions are a guide only.

If the standard product does not fit within your space restrictions or load requirements, custom designs are our speciality and are available upon request. Technofast are continually working on product development and changes may occur at any time. Please contact Technofast for up to date information.

EziJac B3S Modular Spring Return Hydraulic Bolt Tensioner

- Eliminates Pinch Points & Hammering
- Precise Multi-Point Flange Tensioning
- Modular Interchangeable Adaptor Kits
- Automatic Spring Return Mechanism



The **B3S Series EziJac** is a modular hydraulic bolt tensioner featuring an integrated spring return mechanism.

This advanced system automatically resets the tool to its full working stroke after each use, significantly reducing assembly time for critical jointing applications.

Its modular design provides high versatility, allowing multiple adaptor kits to be interchanged with a single load cell. This ensures accurate and repeatable loading while eliminating the pinch points and safety risks associated with traditional power wrenches or hammer tightening.

Operation

The **B3S Series EziJac** uses hydraulic power to tension a bolt to the exact load required, eliminating rotational and torsional stresses. Once the bolt is tensioned, the standard nut is wound down the thread to retain the load.

Upon depressurisation, the spring return mechanism automatically retracts the piston, resetting the tool for the next task. This efficient cycle allows for remote pumping, keeping operators away from the immediate work area.

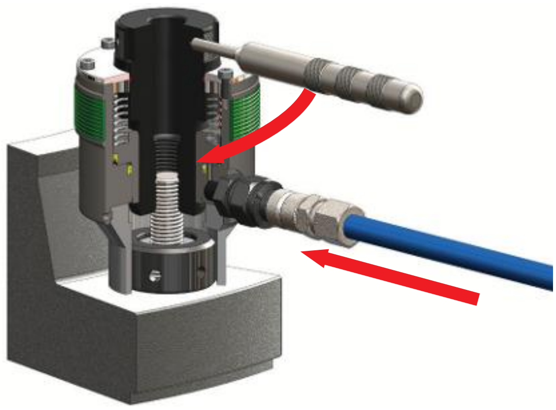
For applications such as flanges, multiple units can be interlinked to provide simultaneous tensioning and even gasket compression.

Features

The **B3S Series EziJac** is designed for high-performance and a long service life in difficult nut locations.

Some features of the **B3S EziJac** are:

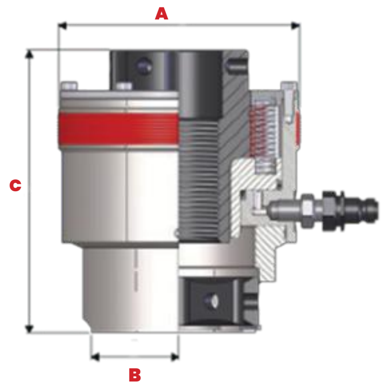
- Modular design supports interchangeable adaptor kits
- Spring return piston provides automatic retraction
- Integrated stroke indicator monitors tool travel
- Nitrile components increase overall tool life
- Leak-free low-friction seals ensure reliable operation
- Long stroke capability suits diverse applications
- Optional second port allows for interconnection
- Compact profile fits confined and difficult spaces

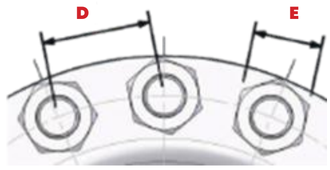


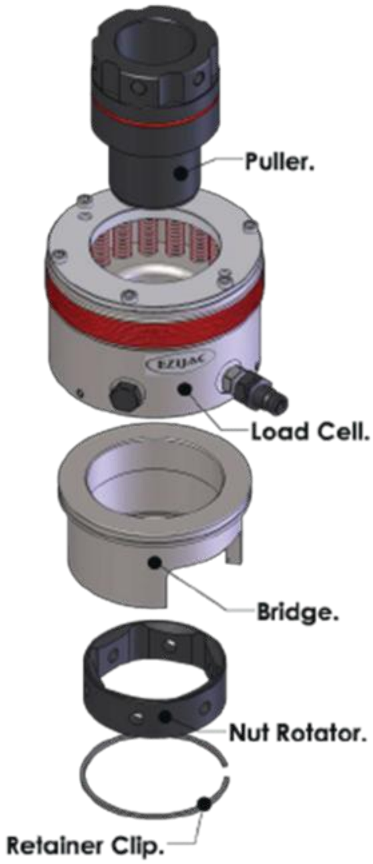
Specifications

Tool #	Size Imp	Size Metric	Kit # (P= TPI pitch)	Max Force (kN)	(A) Total O.D.	(D) Min Pitch	(B) Relief	(C) Total Height	(E) Nut A.F.	Nut Height.	Weight (kg)
	7/8"		SL2-014XP			59	26	122	36.5	22.5	
	1"		SL2-100XP			59	29	122	41.3	25.7	
B3S-L2		M24	SL2-M24X30	361.9	88	59	29	123.5	41	24.2	3.1
	1.1/8"		SL2-102XP			60	32	125	46	28.9	
		M27	SL2-M27X30			60	32	124.5	46	27.6	
		M30	SL3-M30X35			72	36	137.5	50	30.7	
	1.1/4"		SL3-104XP			73	36	140.5	50.8	31.8	
B3S-L3		M33	SL3-M33X35	547.4	105	73	36	142	55	33.7	5
	1.3/8		SL3-106XP			76	39	143.5	55.6	35	
		M36	SL3-M36X40			76	39	144	60	36.6	
	1.1/4"		SL4-104XP			77	36	142	50.6	31.8	
		M33	SL4-M33X35			77	36	146	55	33.7	
	1.3/8"		SL4-106XP	667.6	116	78	39	148.5	55.8	35	
B3S-L4		M36	SL4-M36X40			78	39	149	60	36.6	6.3
	1.1/2"		SL4-108XP			80	42	150	60.3	38.2	
		M39	SL4-M39X40			80	42	150	65	39.6	
	1.1/2"		SL5-108XP			89	42	161.5	60.3	38.2	
		M39	SL5-M39X40			89	42	165	65	39.6	
B3S-L5	1.5/8"		SL5-110XP	965.4	134	91	45	166.5	65.1	41.5	9.2
		M42	SL5-M42X45			91	45	167	70	42	
	1.3/4"		SL5-112XP			94	49	170.5	69.9	44.7	
		M45	SL5-M45X45			94	49	168	75	45	
	1.5/8"		SL6-110XP			99	45	166.5	65.1	41.5	
		M42	SL6-M42X45			99	45	169	70	42	
	1.3/4"		SL6-112XP			101	49	170.5	69.9	44.7	
B3S-L6		M45	SL6-M45X45	1217	149	101	49	172	75	45	11.6
	1.7/8"		SL6-114XP			102	51	173.5	74.6	47.9	
		M48	SL6-M48X50			102	51	178	80	48	
	2"		SL6-200XP			105	54	176.5	79.4	51.1	
		M52	SL6-M52X50			105	54	178	85	52	
	1.7/8"		SL7-114XP			110	51	171.5	74.6	47.9	
		M48	SL7-M48X50			110	51	170	80	48	
B3S-L7	2"		SL7-200XP	1498	162	112	54	174.5	79.4	51.1	14.5
		M52	SL7-M52X50			112	54	177	85	52	
	2.1/4"		SL7-204XP			119	59	180.5	88.9	57.2	
		M56	SL7-M56X55			119	59	180	90	56	
	2.1/4"		SL8-204XP			124	59	185	88.9	57.2	
		M56	SL8-M56X55			124	59	184	90	56	
B3S-L8		M60	SL8-M60X55	1901	184	131	65	187	95	60	20
	2.1/2"		SL8-208XP			131	65	192	98.4	63.6	
		M64	SL8-M64X60			131	65	192	100	64	
	2.1/2"		SL9-208XP			136	65	193.5	98.4	63.6	
		M64	SL9-M64X60			136	65	193	100	64	
B3S-L9		M68	SL9-M68X60	2510	204	146	71	197.5	105	68	26.2
	2.3/4"		SL9-212XP			146	71	200.5	108	70.1	
		M72	SL9-M72X60			146	71	201.5	110	72	
	2.3/4"		SL10-212XP			150	71	203	108		
		M72	SL10-M72X60	2905	219	150	71	205	110	70.1	
B3S-L10	3"		SL10-300XP			157	77	212	117.5	72	31.1
		M76	SL10-M76X60			157	77	209	115	76.5	
	3"		SL11-300XP			161	77	209	117.5	76	
		M76	SL11-M76X60			161	77	209	115	76.5	
		M80	SL11-M80X60			163	77	213	120	76	
	3.1/4"		SL11-304XP	3 687	247	175	87	216	127	80	
B3S-L11		M85	SL11-M85X60			175	87	217	125	82.6	39
	3.1/2"		SL11-308XP			177	93	222	136.5	85	
		M90	SL11-M90X60			177	93	223	135	89.1	
	3.3/4"		SL12-312XP			202	99	232.5	146.1	90	
		M95	SL12-M95X60			202	99	235	145	95.5	
B3S-L12	4"		SL12-400XP	4929	284	212	105	242.5	155.6	95	55.9
		M100	SL12-M100X60			212	105	240	150	102	

If the standard product does not fit within your space restrictions or load requirements, custom designs are our speciality and are available upon request. These dimensions are a guide only.







- Stroke for all Models is 15.0 mm
- Maximum Force is generated at 150 MPa.
- The EziJac is ordered as a complete tool, for example: B3S-L4-M36X40 is specified to suit M36 x 4.
- Adaptor kits are ordered separately, for example SL4-M36X40 is specified to suit M36 X 4. For Imperial Sizes, P = Thread Pitch in Threads Per Inch (TPI).

EziJac C3S Modular Spring Return Hydraulic Bolt Tensioner

- **Eliminates Pinch Points & Hammering**
- **Precise Multi-Point Flange Tensioning**
- **Modular Interchangeable Adaptor Kits**
- **Automatic Spring Return Mechanism**

EZIJAC

The **C3S Series EziJac** is a modular hydraulic bolt tensioner featuring an integrated spring return mechanism.

This advanced system automatically resets the tool to its full working stroke after each use, significantly reducing assembly time for critical jointing applications.

Its modular design provides high versatility, allowing multiple adaptor kits to be interchanged with a single load cell. This ensures accurate and repeatable loading while eliminating the pinch points and safety risks associated with traditional power wrenches or hammer tightening.

Operation

The **C3S Series EziJac** uses hydraulic power to tension a bolt to the exact load required, eliminating rotational and torsional stresses. Once the bolt is tensioned, the standard nut is wound down the thread to retain the load.

Upon depressurisation, the spring return mechanism automatically retracts the piston, resetting the tool for the next task. This efficient cycle allows for remote pumping, keeping operators away from the immediate work area.

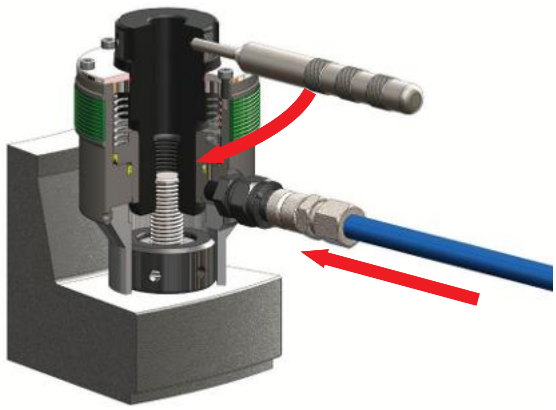
For applications such as flanges, multiple units can be interlinked to provide simultaneous tensioning and even gasket compression.

Features

The **C3S Series EziJac** is designed for high-performance and a long service life in difficult nut locations.

Some features of the **C3S EziJac** are:

- Modular design supports interchangeable adaptor kits
- Spring return piston provides automatic retraction
- Integrated stroke indicator monitors tool travel
- Nitrile components increase overall tool life
- Leak-free low-friction seals ensure reliable operation
- Long stroke capability suits diverse applications
- Optional second port allows for interconnection
- Compact profile fits confined and difficult spaces



Specifications

Tool #	Size Imp	Size Metric	Kit # (P= TPI pitch)	Max Force (kN)	(A) Total O.D.	(D) Min Pitch	(B) Relief	(C) Total Height	(E) Nut A.F.	Nut Height.	Weight (kg)
	7/8"		SL2-014XP			59	26	122	36.5	22.5	
	1"		SL2-100XP			59	29	122	41.3	25.7	
B3S-L2		M24	SL2-M24X30	361.9	88	59	29	123.5	41	24.2	3.1
	1.1/8"		SL2-102XP			60	32	125	46	28.9	
		M27	SL2-M27X30			60	32	124.5	46	27.6	
		M30	SL3-M30X35			72	36	137.5	50	30.7	
	1.1/4"		SL3-104XP			73	36	140.5	50.8	31.8	
B3S-L3		M33	SL3-M33X35	547.4	105	73	36	142	55	33.7	5
	1.3/8		SL3-106XP			76	39	143.5	55.6	35	
		M36	SL3-M36X40			76	39	144	60	36.6	
	1.1/4"		SL4-104XP			77	36	142	50.6	31.8	
		M33	SL4-M33X35			77	36	146	55	33.7	
B3S-L4	1.3/8"		SL4-106XP	667.6	116	78	39	148.5	55.8	35	6.3
		M36	SL4-M36X40			78	39	149	60	36.6	
	1.1/2"		SL4-108XP			80	42	150	60.3	38.2	
		M39	SL4-M39X40			80	42	150	65	39.6	
	1.1/2"		SL5-108XP			89	42	161.5	60.3	38.2	
		M39	SL5-M39X40			89	42	165	65	39.6	
B3S-L5	1.5/8"		SL5-110XP	965.4	134	91	45	166.5	65.1	41.5	9.2
		M42	SL5-M42X45			91	45	167	70	42	
	1.3/4"		SL5-112XP			94	49	170.5	69.9	44.7	
		M45	SL5-M45X45			94	49	168	75	45	
	1.5/8"		SL6-110XP			99	45	166.5	65.1	41.5	
		M42	SL6-M42X45			99	45	169	70	42	
	1.3/4"		SL6-112XP			101	49	170.5	69.9	44.7	
B3S-L6		M45	SL6-M45X45	1217	149	101	49	172	75	45	11.6
	1.7/8"		SL6-114XP			102	51	173.5	74.6	47.9	
		M48	SL6-M48X50			102	51	178	80	48	
	2"		SL6-200XP			105	54	176.5	79.4	51.1	
		M52	SL6-M52X50			105	54	178	85	52	
	1.7/8"		SL7-114XP			110	51	171.5	74.6	47.9	
		M48	SL7-M48X50			110	51	170	80	48	
B3S-L7	2"		SL7-200XP	1498	162	112	54	174.5	79.4	51.1	14.5
		M52	SL7-M52X50			112	54	177	85	52	
	2.1/4"		SL7-204XP			119	59	180.5	88.9	57.2	
		M56	SL7-M56X55			119	59	180	90	56	
	2.1/4"		SL8-204XP			124	59	185	88.9	57.2	
		M56	SL8-M56X55			124	59	184	90	56	
B3S-L8		M60	SL8-M60X55	1901	184	131	65	187	95	60	20
	2.1/2"		SL8-208XP			131	65	192	98.4	63.6	
		M64	SL8-M64X60			131	65	192	100	64	
	2.1/2"		SL9-208XP			136	65	193.5	98.4	63.6	
		M64	SL9-M64X60			136	65	193	100	64	
B3S-L9		M68	SL9-M68X60	2510	204	146	71	197.5	105	68	26.2
	2.3/4"		SL9-212XP			146	71	200.5	108	70.1	
		M72	SL9-M72X60			146	71	201.5	110	72	
	2.3/4"		SL10-212XP			150	71	203	108		
		M72	SL10-M72X60	2905	219	150	71	205	110	70.1	
B3S-L10	3"		SL10-300XP			157	77	212	117.5	72	31.1
		M76	SL10-M76X60			157	77	209	115	76.5	
	3"		SL11-300XP			161	77	209	117.5	76	
		M76	SL11-M76X60			161	77	209	115	76.5	
		M80	SL11-M80X60			163	77	213	120	76	
	3.1/4"		SL11-304XP	3 687	247	175	87	216	127	80	
B3S-L11		M85	SL11-M85X60			175	87	217	125	82.6	39
	3.1/2"		SL11-308XP			177	93	222	136.5	85	
		M90	SL11-M90X60			177	93	223	135	89.1	
	3.3/4"		SL12-312XP			202	99	232.5	146.1	90	
		M95	SL12-M95X60			202	99	235	145	95.5	
B3S-L12	4"		SL12-400XP	4929	284	212	105	242.5	155.6	95	55.9
		M100	SL12-M100X60			212	105	240	150	102	

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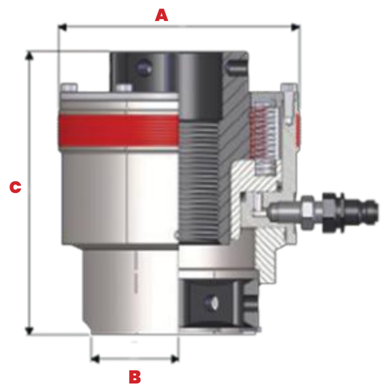
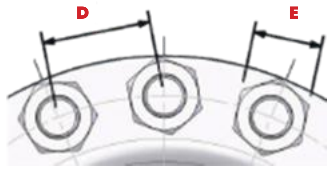
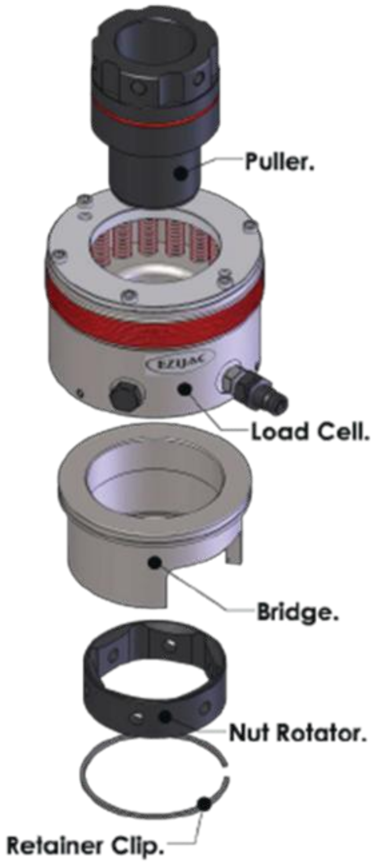






- Stroke for all Models is 15.0 mm
- Maximum Force is generated at 150 MPa.
- The EziJac is ordered as a complete tool, for example: B3S-L4-M36X40 is specified to suit M36 x 4.
- Adaptor kits are ordered separately, for example SL4-M36X40 is specified to suit M36 X 4. For Imperial Sizes, P = Thread Pitch in Threads Per Inch (TPI).

EziJac 2-Stage X2 & Y2 Series Bolt Tensioners

- **Customisable Solutions for Restricted Spaces**
- **High-Force Tensioning in a Slim Footprint**
- **Optional Socket-Driven Nut Gear Drive**
- **Automatic Spring Return Mechanism**

The **EziJac 2-Stage X2 & Y2 Series** offers a versatile range of slim-bodied, multi-stage tensioners specifically engineered for applications with restricted access.

These tools provide high tensioning forces within a compact footprint, allowing for accurate and adaptable loading of bolts and studs.

Designed to meet diverse maintenance requirements, these tensioners enable safe and efficient operation where standard equipment may not fit.

Operation

These hydraulic tensioners provide precise loading without rotational or torsional stress.

An integrated spring return piston automatically resets the tool to full stroke, while over-stroke prevention ensures operational safety.

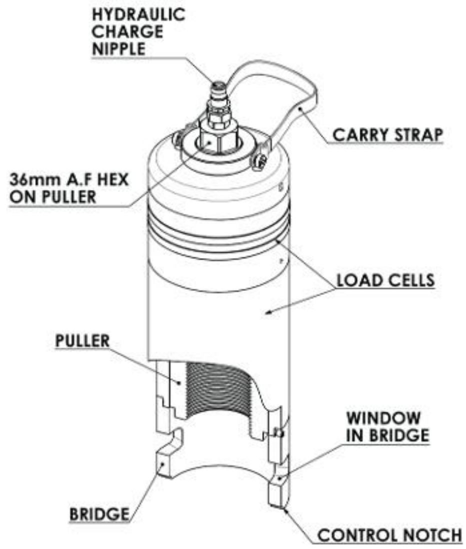
For efficiency in tight locations, an optional side gear drive allows the retaining nut to be rotated using a standard 1/2" socket wrench.

This modular approach ensures the system can be adapted to various bolt sizes and restricted space requirements during simultaneous tensioning proceduressimultaneous tensioning procedures.

Features

The **X2 & Y2 Series** has been designed with the operator in mind, offering customisable configurations to ensure trouble-free operation across diverse applications. Some features of the **X2 & Y2 Series** are:

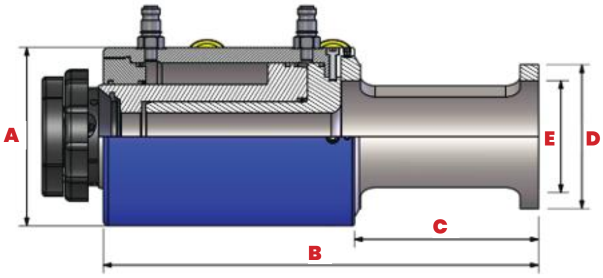
- Slim-fit design for restricted spaces
- Spring return piston for automatic reset
- High tensioning forces in a compact footprint
- Optional gear drive for standard socket wrenches
- Nitride hardening increases tool life
- Leak-free seals ensure reliability
- Stroke indicator for easy monitoring
- Over-stroke prevention for enhanced safety



The **FastaJac Bolt Tensioner** reduces production and maintenance costs for manufacturers and end users of plate heat exchangers. Using lightweight hollow cylinders and the patented Tri-Nut, the system simultaneously compresses plates and gaskets at an even rate, ensuring uniform loading and efficient assembly.

Operation

The Tri-Nut slides over the thread and is activated by a counter-clockwise rotation to engage the bolt. A unique cam lock design keeps the nut secure in the tool, even when positioned upside-down. After tensioning, the action is reversed to lock the Tri-Nut open, allowing the tool to be slid off the bar. This eliminates the need to manually run the nut down the full thread length.



Features

The **FastaJac Bolt Tensioner** has been designed with the operator in mind to ensure a trouble-free and efficient tensioning process. Features include:

- Patented Tri-Nut for rapid installation and removal
- Cam lock design secures tool in any orientation
- Simultaneous compression ensures even gasket loading
- Quick-connect fittings for efficient hydraulic setup
- Safety-first design eliminates operational pinch points
- Lightweight cylinders reduce physical handling effort

Specifications

Load Cell #	Max Pressure (Bar)	Max Force (kN)	Max Stroke	(A)	(B)	(C)	(D)	(E)	Load Cell Weight (kg)	Tri-Nut Size Imperial	Tri-Nut Size Metric	Tri-Nut #
FJ-C27-000	700	403.95	90	132.0	339.5	152.0	120.9	90.0	19.9	1″		C27-I100X8-000
											M24	C27-M24X30-000
										1.1/8″		C27-I102X8-000
											M27	C27-M27X30-000
FJ0C39-000	700	461.5	100	132.0	357.0	157.0	120.0	90.0	20.58		M30	C39-I104X8-000
										1.1/4″		C39-I104X8-000
											M33	C39-M33X35-000
										1.3/8″		C39-I106X8-000
											M36	C39-M36X40-000
										1.1/2″		C39-I108X8-000
											M39	C39-M39X40-000
FJ-C46-000	700	600.7	100	140.0	401.0	192.0	124.5	100.0	21.1	1.5/8″		C46-I110X8-000
											M42	C46-M42X45-000
										1.3/4″		C46-I112X8-000
											M45	C46-M45X45-000
FJ-C52-000	700	639.2	100	154.0	422.5	214.0	140.0	106.0	28.90	1.7/8″		C52-I114X8-000
											M48	C52-M48X50-000
										2.0″		C52-I200X8-000
											M52	C52-M52X50-000


ADVANCE
LIFTING POINTS

LiftaJac

- **High-Force Lifting in a Compact Design**
- **Single or Double-Acting Configurations**
- **Custom Load and Stroke Specifications**
- **Durable High-Strength Alloy Steel**

The **LiftaJac** utilises high hydraulic pressure to generate the significant forces required for lifting large loads. Its compact, high-strength alloy steel design achieves optimal load capacity while maintaining minimal size and weight. Available in single or double-acting, hollow or solid, and short or long-stroke configurations, these jacks are provided in a wide range of sizes and stroke lengths to suit any industrial application.

Operation

The **LiftaJac** is positioned using integrated eyebolts or handles for convenient manoeuvring. Hydraulic pressure is applied to the pressure cylinder, forcing the top cap upward to engage the contact surface and lift the load. For double-acting models, hydraulic pressure is applied to a second nipple to positively retract the **LiftaJac** to its start position. To ensure even lifting of large or complex structures, multiple units can be connected via interlink ports to provide a synchronised operation.

Features

The **LiftaJac** is engineered for durability and safety in demanding lifting environments.

Some features of the Technofast LiftaJac are:

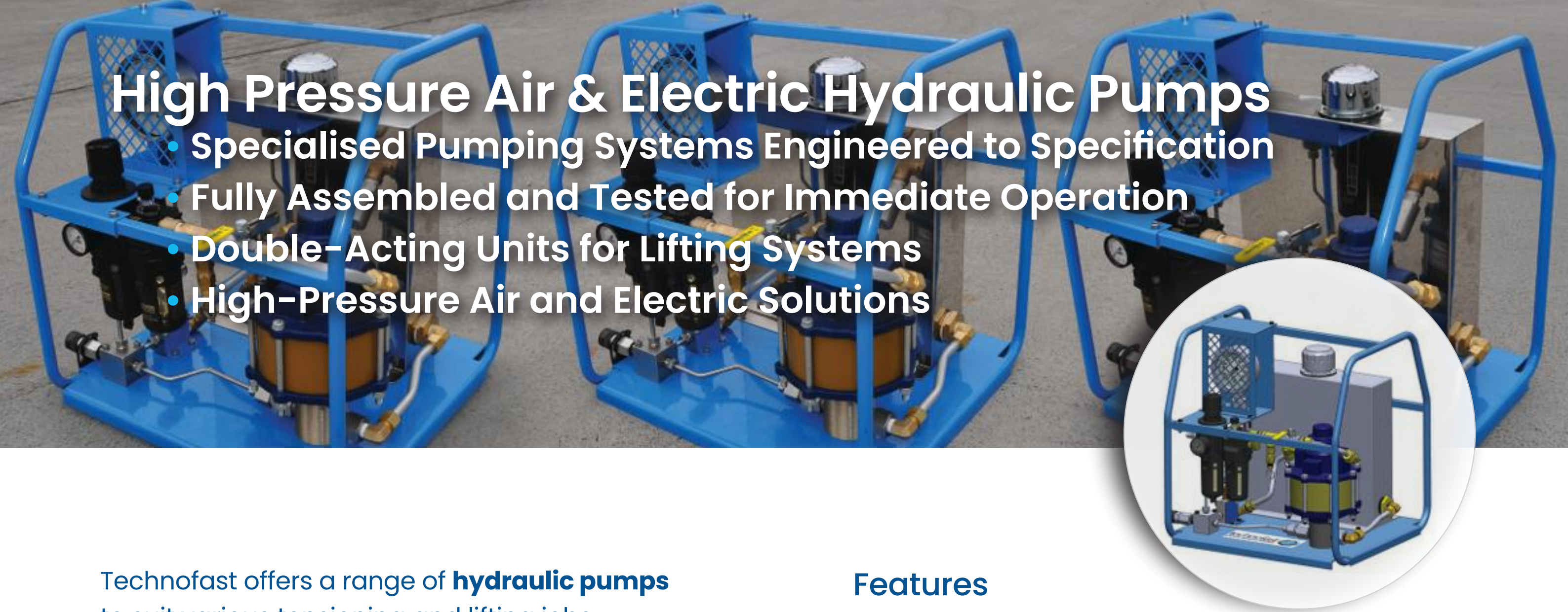
- Available in single or double-acting configurations
- High-strength alloy steel ensures tool durability
- Wide range of sizes, capacities, and stroke lengths
- Removable eye bolts or handles for easy positioning
- Compact design maintains a low weight-to-load ratio
- Optional custom finishes available upon request

1. **Top cap -** Contact surface
2. **Eye bolts -** To assist in lifting the tool to desired location. Removable/side mounted for convenience
3. **Hydraulic nipple 1 -** Hydraulic high pressure inlet
4. **Pressure cylinder** - Cylinder where hydraulic pressure is maintained
5. **Hydraulic nipple 2** - (Double Acting models) Hydraulic pressure inlet to stroke down the lifting jack to its retracted position
6. **Interlink ports** - To operate multiple units simultaneously

Pumps, Hoses and Ancillaries

High Pressure Air & Electric Hydraulic Pumps

- Specialised Pumping Systems Engineered to Specification
- Fully Assembled and Tested for Immediate Operation
- Double-Acting Units for Lifting Systems
- High-Pressure Air and Electric Solutions



Technofast offers a range of **hydraulic pumps** to suit various tensioning and lifting jobs.

All **pumps** are supplied fully assembled and tested ready for operation.

Technofast offers a specialised design service for pumping systems where specific component materials or unique site requirements are required.

This ensures the supply of a pumping solution tailored to your exact application.

Operation

These pumping units are designed to provide hydraulic power for tensioning and lifting applications.

Air-driven models utilise a high-quality pump mechanism to generate pressure, while the electric range provides control via a 240V single-phase motor.

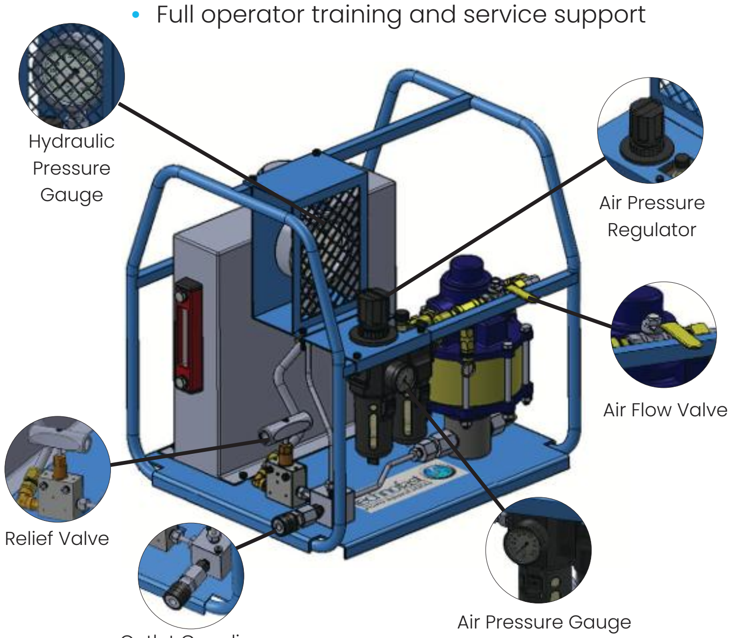
For operation in restricted spaces or to maintain a safe distance, certain models are equipped with a 5-metre remote pendant.

For lifting and clamping systems, double-acting models provide the necessary functionality to both extend and retract cylinders.

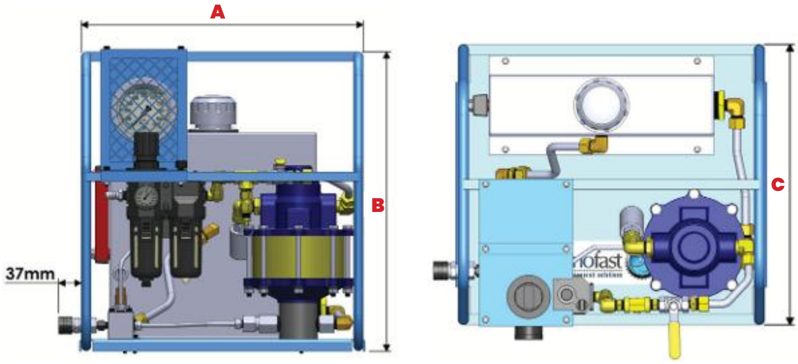
Features

The **hydraulic pump** range has been designed with the operator in mind, offering various features and options for operation. Some features of the range are:

- Fully assembled and operation-tested
- Custom design for specific site needs
- Protective roll cages on featured units
- Precision gauges for pressure monitoring
- Double-acting models for lifting/FastaJac
- Inline lubrication and safety relief included
- Full operator training and service support



Specifications for Air Pumps



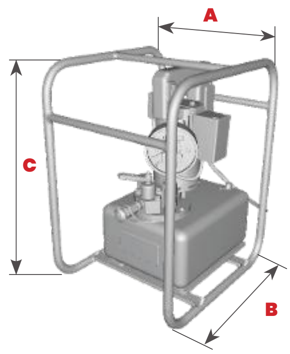
Model #	Working Pressure (Bar)	Low Pressure Flow 0~600 Bar (L/M)	Low Pressure Flow 0~2000 Bar (L/M)	High Pressure Flow 0~2500 Bar (L/M)	High Pressure Flow 2000~4000 Bar (L/M)	Pump Ratio	Reservoir Capacity (L)	Air Consumption (scfm)	Dimensions (mm)			Weight (kg)
									(A)	(B)	(C)	
PU-AH2000H-S	2000	0.3	-	0.15	-	440:1	4	28	410	350	260	17.6
PU-AH2000H	2000	0.3	-	0.15	-	440:1	8	28	464	493	425	31.6
PU-AH2500H	2500	0.6	-	0.3	-	460:1	8	56	464	493	425	31.6
PU-AH4000H	4000	-	0.3	-	0.15	740:1	8	56	464	493	425	31.6
PU-AH2000M	2000	-	-	0.8 max.	-	265:1	25	28	380	650	370	44.5

Other models available by request.

Air Hydraulic Pumps Fitted with:

- 1 x Female Coupler
- Gauge
- Stainless Steel Oil Tanks
- Stainless Steel Air Components
- Powder Coated Roll Cage

Specifications for Electric Pumps

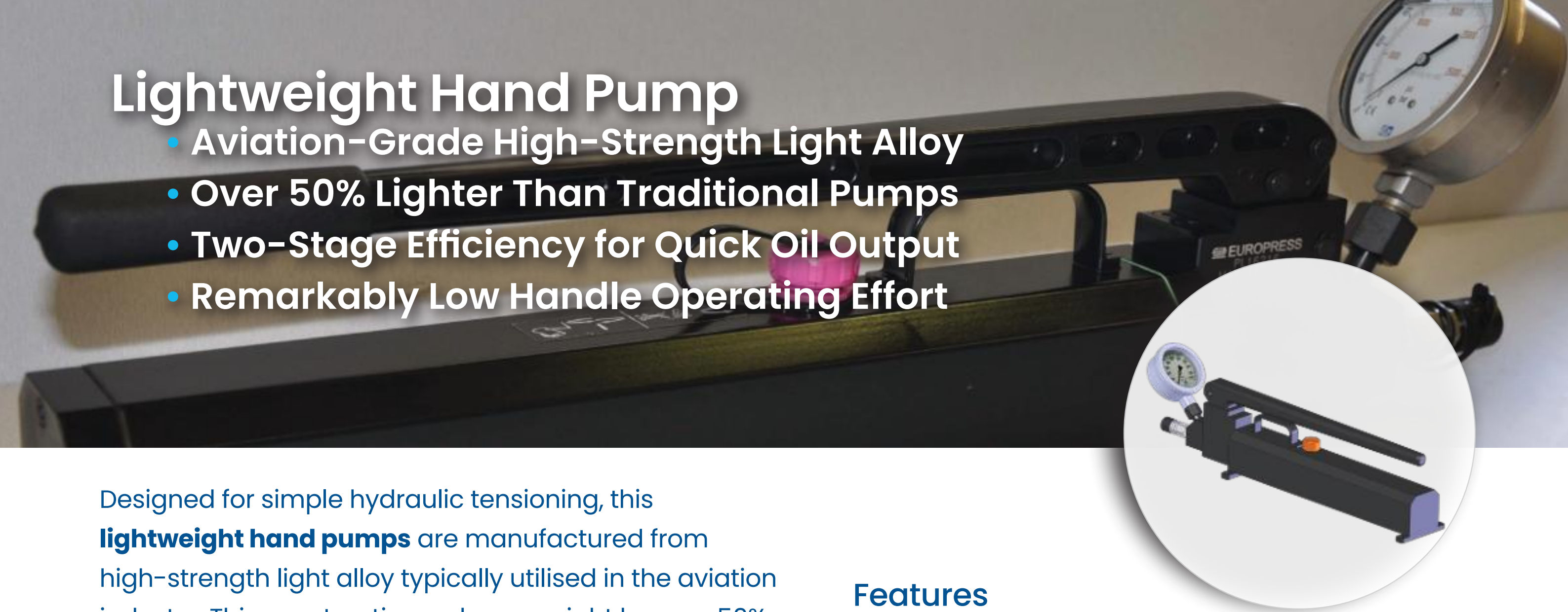


Model #	Pressure 1st Stage (Bar)	Pressure 2nd Stage (Bar)	Flow Rate (L/M)	Reservoir Capacity (L)	Usable Oil Volume (L)	Dimensions (mm)			Weight (kg)
						(A)	(B)	(C)	
PU-EH2000SC	200	2000	0.1-2.0	7	6	325	405	505	26.6

Other models available by request.

240V Electric Hydraulic Pump Fitted with:

- 1 x Female Coupler
- Gauge
- Stainless Steel Roll Cage
- 5mtr Remote Pendant



Lightweight Hand Pump

- Aviation-Grade High-Strength Light Alloy
- Over 50% Lighter Than Traditional Pumps
- Two-Stage Efficiency for Quick Oil Output
- Remarkably Low Handle Operating Effort

Designed for simple hydraulic tensioning, this **lightweight hand pumps** are manufactured from high-strength light alloy typically utilised in the aviation industry. This construction reduces weight by over 50% compared to traditional steel-bodied alternatives, ensuring these durable units are exceptionally easy to handle. Each pump provides high mechanical resistance and is compatible with both single and double-acting cylinders.

Operation

The two-stage cylinder design ensures quick oil output at low pressure and efficient performance at high pressure. An internal safety valve provides automatic relief, while a vacuum oil receptacle and spill-free venting valve allow for efficient operation at any angle. The extended handle remarkably reduces the manual effort required to reach high pressures, saving significant labour during tensioning.

Features

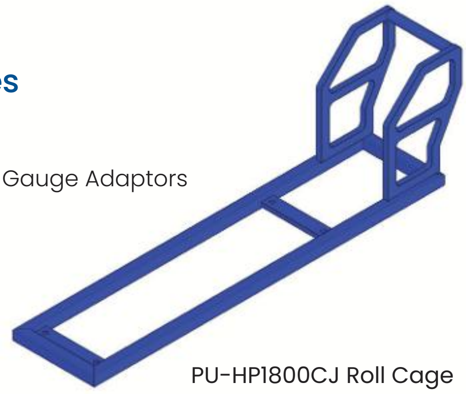
The **Lightweight Hand Pump** has been designed with the operator in mind, offering a range of high-performance features and optional extras to ensure trouble-free operation.

Some features of the **Lightweight Hand Pump** are:

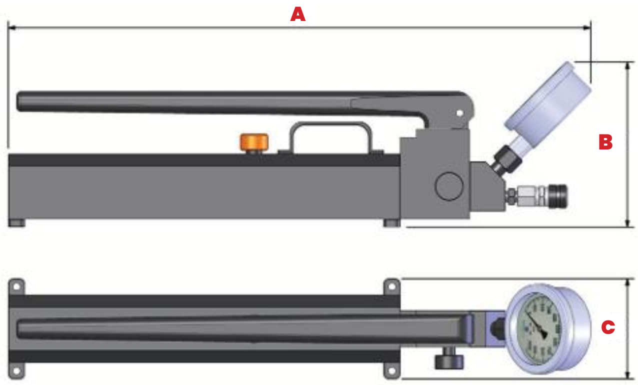
- Adjustable high and low-pressure relief valves
- Vacuum oil receptacle for operation at any angle
- Corrosion-resistant 2000ml stainless steel tank
- Fitted with original CEJN hoses and adaptors
- Integrated carry handle and mounting holes
- Additional models available upon request
- Operator training is available and all pump servicing fully supported by Technofast

Accessories

- Roll Cage
- Gauges and Gauge Adaptors
- Fittings
- Spare parts



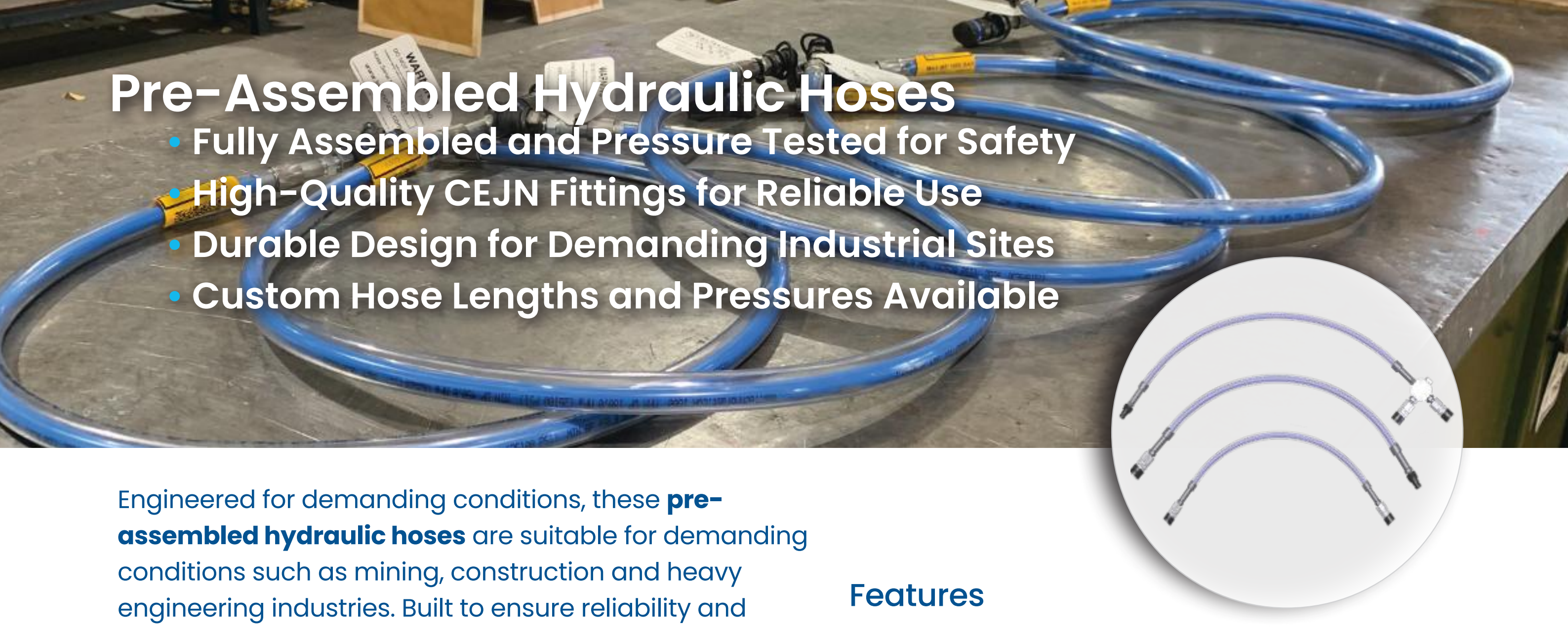
PU-HP1800CJ Roll Cage



Specifications

Model #	Pressure 1st Stage (Bar)	Pressure 2nd Stage (Bar)	Handle Effort (N)	Reservoir Capacity (L)	Usable oil Volume (L)	Dimensions (mm)			Weight (kg)
						(A)	(B)	(C)	
PU-SHP700C	-	700	363	1.3	1.1	-	-	123	5
PU-HP700C	-	700	363	2.3	1.9	-	-	123	6.4
PU-HP1800CJ	20	1800	522	2.2	1.8	695	266	125	10

Other models available by request.



Pre-Assembled Hydraulic Hoses

- Fully Assembled and Pressure Tested for Safety
- High-Quality CEJN Fittings for Reliable Use
- Durable Design for Demanding Industrial Sites
- Custom Hose Lengths and Pressures Available

Engineered for demanding conditions, these **pre-assembled hydraulic hoses** are suitable for demanding conditions such as mining, construction and heavy engineering industries. Built to ensure reliability and precision, every hose is supplied fully assembled and pressure-tested for immediate operation.

Operation

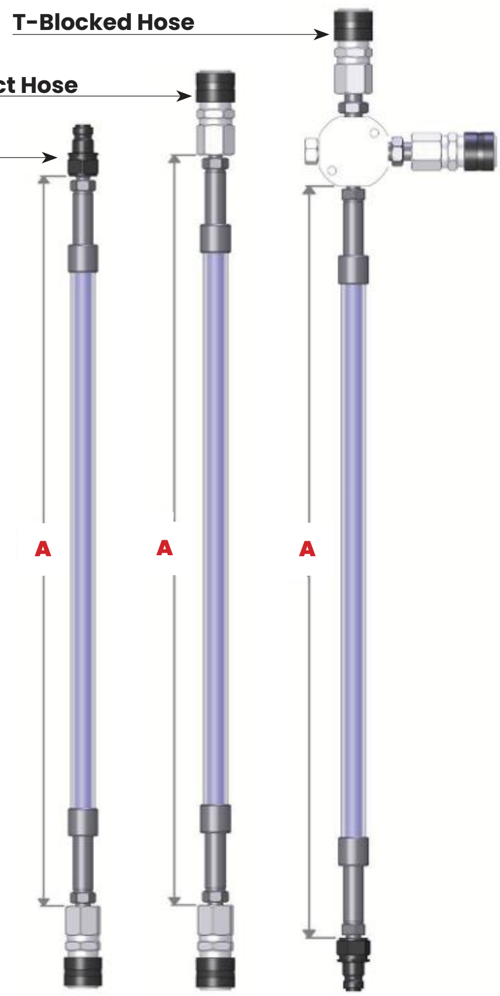
These **pre-assembled hydraulic hoses** are designed to provide secure, high-pressure connections between pumping units and tensioning tools. Available in multiple configurations, they allow for flexible hydraulic setups, whether connecting a single tool or interlinking multiple units for simultaneous operation. The use of high-quality fittings ensures leak-free performance and ease of connection, even in tough industrial environments. Longer lengths or higher-pressure hoses can be manufactured to meet specific project requirements.

Features

The **pre-assembled hydraulic hose** range has been designed with the operator in mind, offering various configurations for trouble-free operation.

Some features of the range are:

- Supplied fully assembled & tested for immediate use
- Durable construction engineered for tough industrial conditions
- Custom lengths and higher pressure ratings available on request
- Optimised designs suitable for diverse hydraulic setups



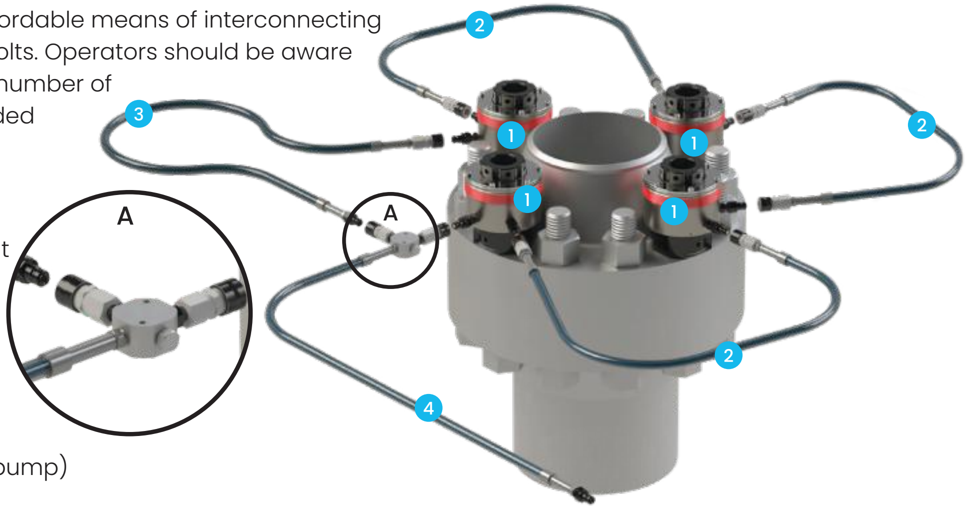
Specifications

Hose Type	Part #	Max Working Pressure (Bar)	Min Burst Pressure (Bar)	(A) Hose length (m)	Min Bend Radius (mm)	CEJN Couplers Fitted
LINK HOSE	PU-HG090L	1500	3000	0.9	130	1 x Male & 1 x Female
	PU-HG150L	1500	3000	1.5	130	
	PU-HG300L	1500	3000	3.0	130	
	PU-HG400L	1500	3000	4.0	130	
	PU-HG500L	1500	3000	5.0	130	
	PU-HG800L	1500	3000	8.0	130	
INTER-CONNECT HOSE	PU-HG090	1500	3000	0.9	130	2 x Female
	PU-HG150	1500	3000	1.5	130	
	PU-HG300	1500	3000	3.0	130	
	PU-HG400	1500	3000	4.0	130	
	PU-HG500	1500	3000	5.0	130	
	PU-HG800	1500	3000	8.0	130	
T-BLOCKD HOSE	PU-HG090TB	1500	3000	0.9	130	1 x Male & 2 x Female
	PU-HG150TB	1500	3000	1.5	130	
	PU-HG300TB	1500	3000	3.0	130	
	PU-HG400TB	1500	3000	4.0	130	
	PU-HG500TB	1500	3000	5.0	130	
	PU-HG800TB	1500	3000	8.0	130	

Hose Setup Guide

Daisy Chain

The Daisy chain hose package is an affordable means of interconnecting multiple Tensioners, Hydraulic Nuts or Bolts. Operators should be aware that as return oil must pass through a number of Tensioners, this layout may have extended reset times for Spring Return tools.

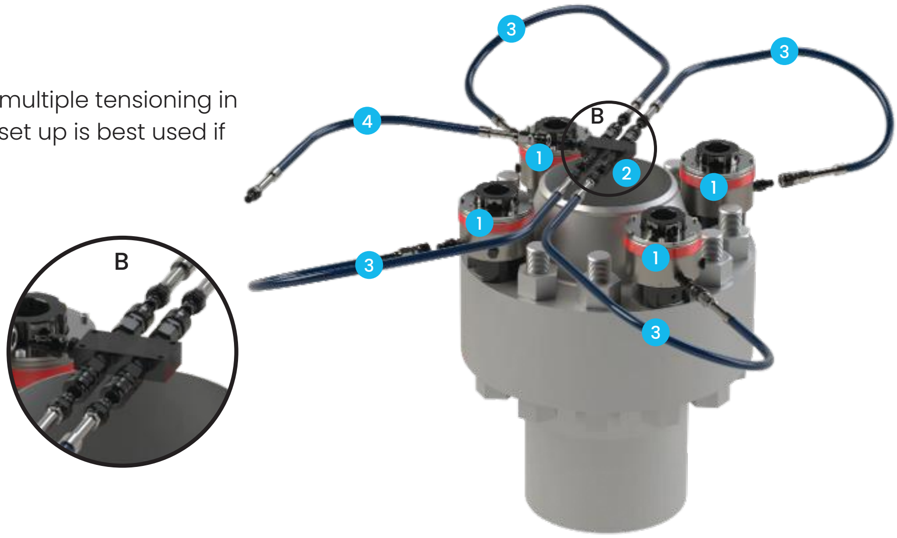


Daisy Chain Components

1. Bolt tensioner or hydraulic nut/bolt (Additional nipple supplied)
2. Interconnect hose
3. Link hose (Linking the loop)
4. T-Blocked hose (Feed hose from pump)

Manifold

The manifold hose package is suited for multiple tensioning in groups of 4 to 8 or more tensioners. This set up is best used if there are grouped tools to tension.

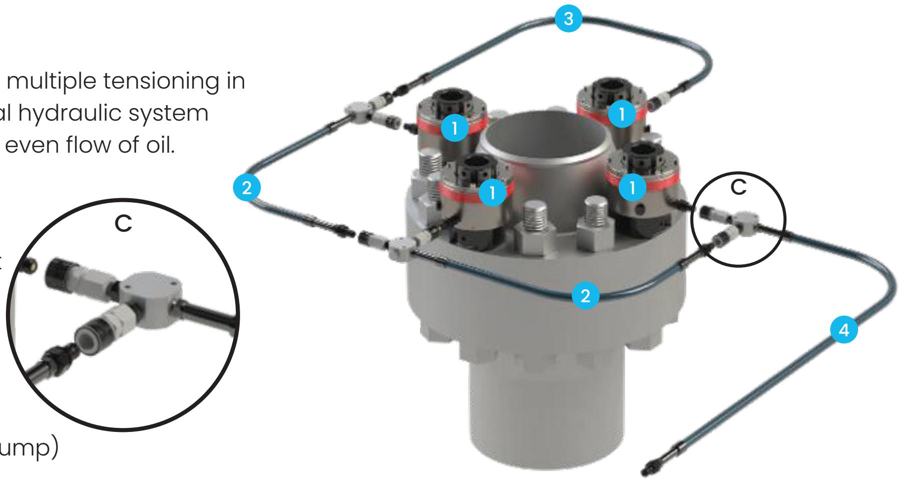


Manifold Components

1. Bolt tensioner or hydraulic nut/bolt
2. 5-Way manifold
3. Link hose
4. Link hose (Feed hose from pump)

T-Block

The T-Blocked hose package is suited for multiple tensioning in large groups. This set up is a 100% external hydraulic system allowing for quick tool return speeds and even flow of oil.



T-Block Components

1. Bolt tensioner or Hydraulic nut/bolt
2. T-Blocked hose
3. Link hose
4. T-Blocked hose (Feed hose from pump)

Ancillary Items

C-Spanners

Part #	Description
Walter 68/75	C-Spanner to Suit Dia. 68mm to 75mm
Walter 80/90	C-Spanner to Suit Dia. 80mm to 90mm
Walter 95/100	C-Spanner to Suit Dia. 95mm to 100mm
Walter 110/115	C-Spanner to Suit Dia. 110mm to 115mm
Walter 135/145	C-Spanner to Suit Dia. 135mm to 145mm
Walter 155/165	C-Spanner to Suit Dia. 155mm to 165mm
Walter 180/195	C-Spanner to Suit Dia. 180mm to 195mm
Walter 205/220	C-Spanner to Suit Dia. 205mm to 220mm

Pin Spanners

Part #	Description
TB-06X150-000	Pin Spanner to suit Dia. 6mm hole
TB-08X160-000	Pin Spanner to suit Dia. 8mm hole
TB-10X170-000	Pin Spanner to suit Dia. 10mm hole
TB-12X170-000	Pin Spanner to suit Dia. 12mm hole
TB-14X170-000	Pin Spanner to suit Dia. 14mm hole

Hydraulic Fittings

Part #	Description
HNCP001	Std 1/8" BSP Male 116 Series Nipple (10 116 6281)
HNCP002	Std Female 116 Series Snap Coupler (10 116 1202)
HNCP003	Male Nipple to 1/4" Female port
HNCP011	Nipple Extension Adaptor
HNCP012	1/8" BSP to M8 Cone Seat Nipple Extension
HNCP018	1/8" BSP Grub Screw
HNCP019	Adaptor M10x1.25 -1/4" BSP Nipple Adaptor
HNCP022	1/8" Blanking Plug
HNCP028	1/4" male TO 1/4" BSP Male Nipple Extension
HNCP029	Banjo Fitting 1/8" BSP with Dowty Seals
HNCP043	1/8" BSP Dowty Seal Plug Recessed
HNCP069	Modified HNCP001 Nipple 1.9 mm Shorter Stem
HNCP094	1/4" Blanking Plug to suit 4-Way Manifold
HNCP117	90 degree 116 Series Swivel Coupler
HNCP190	1/8" BSP to M6 Cone Seat Nipple Extension
HNCP212	T-Block (c/w 1 Male & 2 Female Fittings)
HNCP214	5 Way Manifold (c/w 1 Male & 4 Female Fittings)

HNCP002 - Std Female 116 Series Snap Coupler (10 116 1202)

HNCP028 - 1/4" male TO 1/4" BSP Male Nipple Extension

HNCP214 - 5 Way Manifold (Fitted with 1 x Male & 4 x Female Fittings)

Specialty Services

General Servicing & Hose Pressure Testing Service

To support the continued safety and performance of your equipment, a range of specialised maintenance and testing services is available for all hydraulic bolt tensioning systems.

These services ensure that tools and ancillary components are maintained to the highest quality through rigorous inspection and testing procedures.

By adhering to strict performance standards, these preventative measures help to identify potential faults, manage tool lifecycles, and improve overall site safety.

Tooling Maintenance and Servicing

General servicing is offered for all hydraulic tensioning tooling, including hydraulic nuts, bolt tensioners, and various ancillary equipment.

Many modern designs incorporate highly stressed components with a finite usage cycle life; therefore, correct operating procedures and regular testing are essential.

During service, all tooling is completely stripped and undergoes a comprehensive assessment by production and engineering teams.

Detailed finding, testing, and completed service reports are then generated for your records.

Hose Pressure Testing and Certification

Flexible hoses can sustain damage from being stretched, twisted, crushed, or coiled below their minimum bend radius, which can lead to leaks or catastrophic failure.

In accordance with ISO 6605 and SAE J343 standards, high-pressure hydraulic hose assemblies used in heavy industry require annual testing and recertification.

A complete testing, certification, and identification service is provided to ensure the integrity of your equipment and reduce risk to the user.

Technofast Hose Service includes:

- Full visual inspection
- Thorough Clean and Oil Replacement
- Stringent Pressure Testing to above maximum working pressure
- Test Tag installed and updated stating the next test due date
- Provision of a Technofast Serial Number allowing the full working history of the Hose to be recorded for future reference
- An ongoing Maintenance Record retained by Technofast
- Failure Reports generated as required

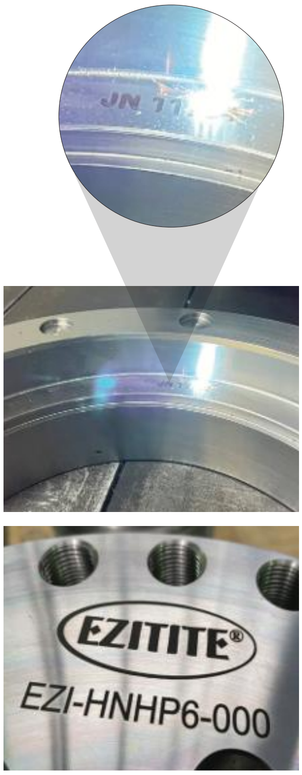
Industrial Laser Marking

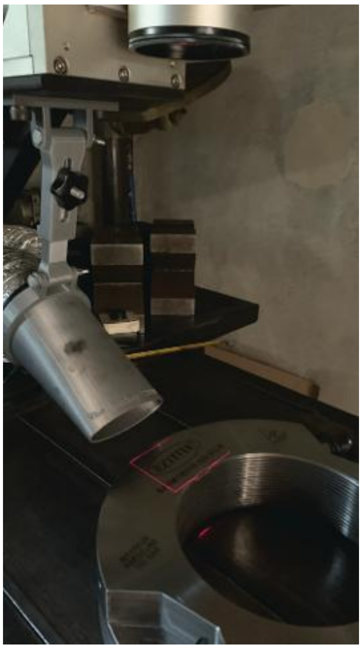
Technofast offers a comprehensive marking service utilising state-of-the-art, computer-operated technology. This system guarantees precise identification without the use of chemicals or ink. Currently supplying major electrical distribution and transportation companies across Australia, our services support everything from individual labels and logos to barcode and QR code marking. Every project is completed to the highest standard, ensuring durability and long-term traceability for components across the engineering, construction and mining sectors.

Advanced Laser Systems

The process uses a digitally controlled, focused beam to alter a material's surface properties without the need for inks or coatings. Depending on the application, we utilise Fiber, CO^2 or Galvo lasers to achieve surface colour changes, controlled engraving or micro-structuring.

These systems integrate into existing manufacturing processes to reduce downtime, while MOPA technology ensures fine control over finish and contrast. Marks are placed with extreme accuracy, ensuring they remain readable throughout the product's entire lifecycle.





Materials and Applications

Our machines are configured to mark a wide range of industrial materials including aluminium, stainless steel, titanium, carbon fibre and polymers. Marking tools and equipment improves accountability and safety through the application of:

- **Identification:** Product, part, job and serial numbers, including Australian Made references
- **Data Capture:** High-contrast QR codes, barcodes and data matrix codes optimised for asset tracking
- **Traceability:** Permanent marks for product authentication, quality assurance and maintenance tracking

Engraving vs Marking

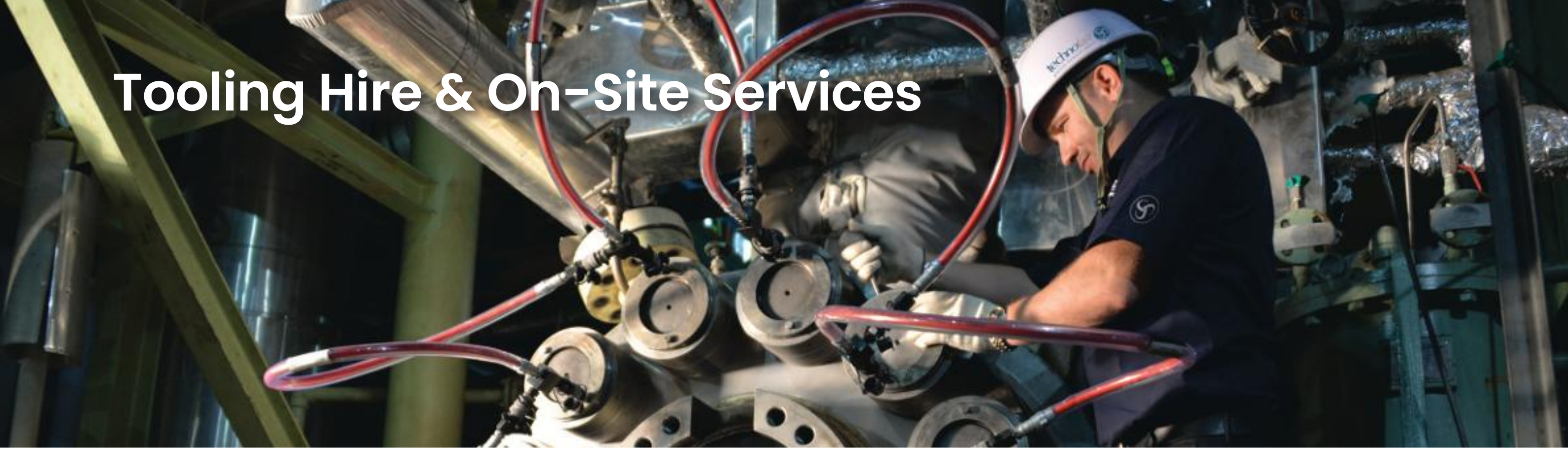
While the terms are often used interchangeably, they serve different industrial purposes:

- **Laser Engraving:** physically removes material to form deeper, permanent marks that are both visible and tactile. It is ideal for parts enduring extreme industrial wear
- **Laser Marking:** Alters the surface contrast through oxidation without removing material. This maintains part integrity, providing smooth, high-contrast results for barcodes and logos

Technofast Laser Capabilities

- **Open Room Laser:** No size limitations for large tooling or equipment
- **4th Axis Rotation:** Handles components up to 300.0mm diameter and 20.0kg
- **Marking Area:** 260.0mm x 260.0mm in a single pass (larger marks via multiple passes)
- **Versatile Scale:** Suitable for tiny components through to large asset registers

Tooling Hire & On-Site Services



Technofast maintains an extensive fleet of **hire tools** across our Queensland and Western Australia facilities, ensuring equipment is readily accessible for national projects.

This exclusive range is available for one-off service works or continued projects, providing a cost-effective solution for those who require precision hydraulic bolt tensioning without long-term capital investment.

Hire also offers the opportunity to familiarise operators with specific tooling before committing to a purchase.

Hire Options and Support

Packages are fully tailorable to meet specific project requirements, with various hire periods available.

Equipment-Only Hire: Tools are supplied fully tested and ready for site use. This option is ideal for self-managed projects, with the additional support of our team for initial on-site setup and installation if required

Operated Equipment Service: This service provides access to a team of highly skilled operators who bring quality experience and on-site problem-solving to every job. We recommend this option to ensure technical expertise is available throughout the project

On-Site Installation & Technical Support: Technofast operators are available to complete installations and perform necessary bolt load and flange stress calculations.

Custom Requirements

Supported by an expert engineering team and purpose-built facilities, we can meet custom tool requirements for unusual sizes or applications.

This includes the design and development of new, specialised tools or the creation of custom spare parts to repair existing equipment.

Equipment Fleet

Our specialised **hire** solutions include:

- EziJac Hydraulic Bolt Tensioners
- Hydraulic Torque Wrenches
- Flange Facing and X-Y Milling Machines
- Complete Pump, Hose and Fitting Packages

Product Demonstrations & Equipment Training

To ensure the safety, mechanical integrity, and productivity of your investment, Technofast provides comprehensive **training** for all hydraulic systems and processes.

These tailored packages are designed to familiarise end users with the specific operation and maintenance of their equipment.

By incorporating site-specific procedures and relevant requirements, we ensure your team achieves the maximum optimisation from every Technofast product.

Operational Training Delivery

Training is available for both small and large groups through various flexible delivery methods:

On-Site Programs: Conducted at your facility to provide hands-on experience during or prior to equipment installation

Virtual Workshops: Digital training sessions designed for remote teams or theoretical overviews

Engineering Demonstrations: Complex technical sessions and demonstrations held at our Technofast Head Office in Crestmead, Queensland

Training Program Features

Our **training** program ensures an exceptional level of quality and consistency. Each session covers:

- **Comprehensive Tool Review:** In-depth guidance on the correct use of our exclusive tooling range
- **Safety and Maintenance:** Essential knowledge regarding safety precautions and preventative equipment care
- **Tailored Content:** Presentations adjusted to suit the existing skill sets and knowledge of your operators
- **Certification:** Provision of reference materials and registered completion of training documents for all attendees

Precision Engineering & Design Services

Technofast operates a well-equipped facility featuring advanced technology and a highly skilled workforce dedicated to producing **precision-engineered** solutions.

By conducting all aspects of **design, machining, and testing** in-house, we maintain full control over quality and delivery timelines.

Our commitment to continuous improvement ensures that we consistently meet the stringent standards of third-party certifying organisations while providing exceptional service with meticulous attention to detail.

Engineering and Custom Design

Our engineering staff utilise state-of-the-art 3D modelling and drafting packages to develop our range of unique hydraulic tensioning solutions.

- **Performance Optimisation:** Designs are assessed using Finite Element Analysis (FEA) to run virtual experiments. This allows our team to optimise custom solutions, reduce component size, and achieve significant cost savings for clients
- **3D Printing and Prototyping:** We utilise 3D printing technology to verify fitment and feel on specific applications. This allows users to confirm a proposed custom solution before moving to manufacture
- **Specialised Engineering:** Our team handles diverse tasks from bespoke nuts and bolts to the production of intricate prototype parts

In-House Manufacturing and Testing

All components are produced in-house using modern CNC machines to ensure exceptional quality throughout the manufacturing process.

- **CNC Milling and Turning:** Versatile machining capabilities for both custom components and small, complex pieces
- **Materials and Finishing:** Certified alloy steels are typically used for maximum toughness. Products can be finished with various surface treatments to suit specific environmental conditions
- **Testing and Validation:** Rigorous load testing and cycle testing are performed to ensure mechanical integrity and reliability
- **Dot Peen Marking:** Permanent mechanical identification is available for asset registers, including part numbers, serial numbers, and special instructions

Technofast Services - WA

Technofast Services, Rockingham WA

Technofast Services is a dedicated division specialising in comprehensive on-site support for tensioning, machining and infrastructure maintenance.

Based in Rockingham, Western Australia, since 1992, our 100% Australian owned and operated division has successfully delivered major projects for leading industrial clients across Australia and the Asian region.

We provide competitive, custom-designed solutions with a focus on prompt supply and technical excellence.

On-Site Capabilities

Our skilled operators are available to perform professional installations and technical services across a wide range of industrial applications

- **Precision Machining:** Flange facing, drilling and hot tapping
- **Specialised Bolting:** Bolt tensioning, torque tensioning and subsea bolting operations
- **Asset Maintenance:** Leak sealing and on-site installation of Technofast products
- **Technical Analysis:** Comprehensive bolt load and flange stress calculations
- **Safety Training:** On-site staff training for the safe operation of hydraulic tensioning devices

Specialised Hire Fleet

A comprehensive range of bolt tensioning and machining equipment is available for short-term or extended hire periods to support your project requirements.

Equipment including:

- **Tensioning and Torque:** Hydraulic bolt tensioners and torque wrenches
- **System Support:** High-pressure pumps, hoses and all necessary fittings
- **Portable Machining:** Flange facing machines and X-Y milling units

Contact Details

For a quote or further information, please contact our Western Australian team:

Technofast Services - WA
Unit 3, 15 Savery Way, Rockingham WA 6168
+61 409 295 776
saleswa@technofast.com

More Information

Enquiry Form:

To help us provide a prompt and accurate recommendation, our consultants need specific details to identify the right product for your needs. To assist this process, please copy this page and complete the following data and send your information to: **sales@technofast.com**

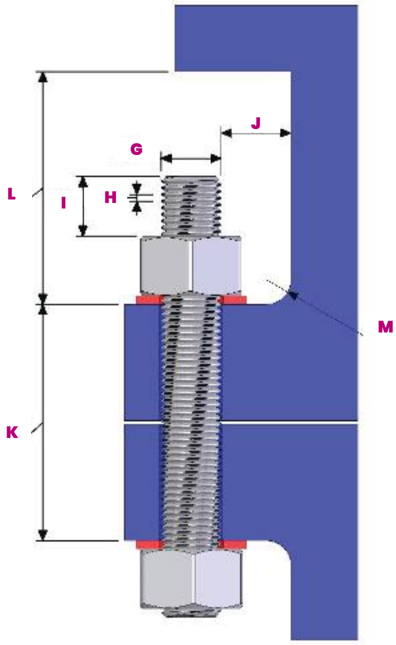
*Fields marked with * are required*

Completed By	
Company	
Date	
Bolt Material	
Required Bold Load*	
Torque or Force	

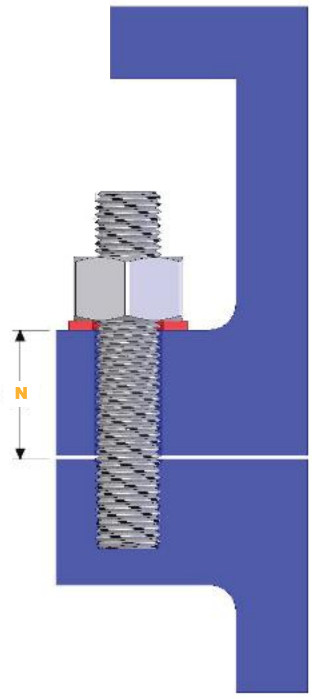
A*	Nut height	
B*	Nut across flats	
C	Nut across corners	
D	Washer Ø	
E	Washer thickness	
F*	Bolt centres	
G*	Thread size	
H	Thread pitch	
I	Min.	
	Max.	
J	Edge of bolt to obstruction	
K	Through bolt	
L	Top clearance	
M	Weld or redius details	
N	Blind bolt	

Comments

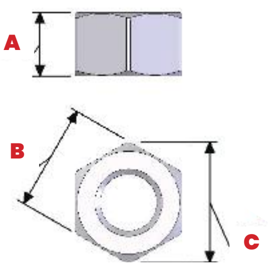
Through Bolt



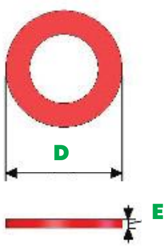
Blind Bolt



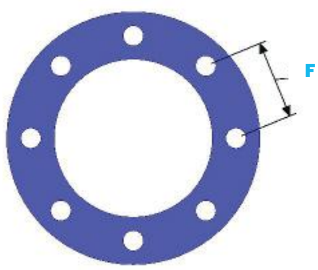
Nut Details



Washer Details



Flange Details



Notes

technofast
precision engineered solutions

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